



BANARAS HINDU UNIVERSITY

B.Sc.(Hons.) Courses

Effective from 2024 under NEP

Offered by

Faculty of Science



BANARAS HINDU UNIVERSITY



Department of Geology

Faculty of Science

SYLLABI

B.Sc. (Hons.) Programme

In

Geology

COURSE STRUCTURE FOR 4-YEAR UG GEOLOGY, NEP-2020 Department of Geology, Banaras Hindu University		
Semester	Courses	Credits
I	ESMJ11: Elements of Earth Sciences	3
	ESMJ12: Practical connected with ESMJ11	1
	ESMD11: Evolving Earth	3
	ESSE11: Smartphone Geosciences	2
	ESSE12: Practical connected with ESSE11	1
	Minor same as major ESMJ11, ESMJ12	3+1
	VA (To be offered by the University)	2+2
	AE (To be offered by the University)	2
	Total Credits	20
II	ESMJ21: Minerals and Earth Material	3
	ESMJ22: Practical connected with ESMJ21	1
	ESMD21: Earth, Environment and Society	3
	ESSE21: Gemology	2
	ESSE22: Practical connected with ESSE21	1
	Minor same as major ESMJ21, ESMJ22	3+1
	VA (To be offered by the University)	2+2
	AE (To be offered by the University)	2
	Total Credits	20
III	ESMJ31: Physical Geology	3
	ESMJ32: Elementary Structural Geology	3
	ESMJ33: Practical connected with ESMJ31	1
	ESMJ34: Practical connected with ESMJ32	1
	ESMD31: Rocks, Minerals and Fossils	3
	ESSE31: Geological Field Training	3
	ESMV31: Minerals and Earth Material*	3
	ESMV32: Practical connected with ESMV31*	1
	AE (To be offered by the University)	2
	<i>* Students may opt any four-credit course from Online SWAYAM courses approved by the Department.</i>	
	Total Credits	20
IV	ESMJ41: Petrology and Economic Geology	3
	ESMJ42: Paleontology and Stratigraphy	3
	ESMJ43: Environmental Geology and Hydrogeology	3
	ESMJ44: Essentials of Remote Sensing	2
	ESMJ45: Practical connected with ESMJ41	1
	ESMJ46: Practical connected with ESMJ42	1
	ESMJ47: Practical connected with ESMJ43	1
	ES-Minor: same as major ESMJ43 and ESMJ47*	4
	AE (To be offered by the University)	2
	<i>* Students may opt any four credit course from Online SWAYAM courses approved by the Department.</i>	
	Total Credits	20

V	ESMJ51: Metamorphic Petrology	3
	ESMJ52: Sedimentary Petrology	3
	ESMJ53: Igneous Petrology	3
	ESMJ54: Mineralogy and Crystallography	3
	ESMJ55: Practical connected with ESMJ -51 and ESMJ -52	1
	ESMJ 56: Practical connected with ESMJ -53 and ESMJ -54	1
	ESMV51: Geoarcheology and Museology *	2+2
	ESIN-51: (Internship): [Students can do internship after first to fourth semester during summer vacation]	2
	<i>*Students may opt any four-credit course from Online SWAYAM courses approved by the Department.</i>	
	Total Credits	20
VI	ESMJ61: Structural Geology	3
	ESMJ62: Economic Geology	3
	ESMJ63: Paleontology	3
	ESMJ64: Stratigraphy	3
	ESMJ65: Practical connected with ESMJ61 and ESMJ62	1
	ESMJ66: Practical connected with ESMJ63 and ESMJ64	1
	ESMJ67: Field Geology	2
	ESMV61: Agriculture Geology *	2+2
	<i>*Students may opt any four credit course from Online SWAYAM courses approved by the Department.</i>	
	Total Credits	20
VII-H	<i>Any four major elective courses from ESMJ71 to ESMJ75 form the list with connected practical (3 credit each)</i>	4x3
	ESMJ71: Geomorphology and Geotectonics	2
	ESMJ72: Foundations of Engineering Geology and Surveying	2
	ESMJ73: Geochemistry and Instrumentation	2
	ESMJ74: Exploration Geosciences	2
	ESMJ75: Fuel Geology	2
	ESMJ76: Practical connected with ESMJ71	1
	ESMJ77: Practical connected with ESMJ72	1
	ESMJ78: Practical connected with ESMJ73	1
	ESMJ79: Practical connected with ESMJ74	1
	ESMJ710: Practical connected with ESMJ75	1
	ESMJ711: Advanced Geological Field Training	4
	ES-Minor: same as major ESMJ71 and ESMJ75*	4
	<i>* Students may opt any four credit course from Online SWAYAM courses approved by the Department.</i>	
	Total Credits	20

VIII-H	<i>Any four major elective courses from ESMJ81 to ESMJ85 along with connected practical (4 credit each)</i>	4x4
	ESMJ81: Marine Geology, Oceanography and Climate	3
	ESMJ82: Georesource and Mineral Policy	3
	ESMJ83: Ground Water Hydrology and Soil Geology	3
	ESMJ84: Remote Sensing and GIS	3
	ESMJ85: Basin Analysis	3
	ESMJ86: Practical connected with ESMJ81	1
	ESMJ87: Practical connected with ESMJ82	1
	ESMJ88: Practical connected with ESMJ83	1
	ESMJ89: Practical connected with ESMJ84	1
	ESMJ810: Practical connected with ESMJ85	1
	ES-Minor: same as major ESMJ84 and ESMJ89*	3+1
	<i>* Students may opt any four credit course from Online SWAYAM courses approved by the Department.</i>	
	Total Credits	20
	Research Students	
VII-R	<i>Same as VII-H</i>	20
VIII-R	<i>Any one major elective courses from ESMJ81 to ESMJ85 along with connected practical (4 credit each)(List Same as VIII-H)</i>	(3+1)
	ES-Minor: same as major ESMJ71 and ESMJ75*	4
	ES-DS8R1: Dissertation	12
	<i>* Students may opt any four credit course from Online SWAYAM courses approved by the Department.</i>	
	Total Credits	20

SEC – Skill Enhancement Course;
AEC – Ability Enhancement Course;
VAC – Value Added Course;
MD – Multidisciplinary Course

1 Credit (Theory) = 15 Hours

1 Credit (Practical) = 30 Hours

Semester: I

Course Title	ESMJ11: Elements of Earth Sciences			
Category of Course	Major			
Credits& Hour of Teaching		Theory	Practical	Cumulative
	Credits	03	----	03
	Hour of Teaching (Total)	45	----	45
Course Objectives (not more than 100 words)	The objective of this paper is to provide students with a foundational understanding of Earth as a whole and its place in the solar system, while also introducing the geological time scale and structural geology concepts.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Earth as a planet, its origin, size, shape, mass, density, rotational and revolutional parameters. The internal constitution of the earth, core, mantle, and crust; Overall composition of the Earth and comparison to other planetary bodies.			10
II	Catastrophism, Neptunism, Plutonism, Uniformitarianism, law of superposition, law of faunal succession, and law of original horizontality. Origin and evolution of atmosphere, hydrosphere and biosphere. Weathering, Erosion, and Mass wasting- processes and types.			10
III	Earthquakes & distribution of earthquake belts; Volcanoes, its types& distribution of volcanic belts on Earth; Geological time scale; Major events in the earth's history through Geological time, Age of the Earth & basic idea on dating.			10
IV	Introduction to planar and linear structures; Identification of primary structures, and secondary structures, folds, faults, and joints. Measurements of the orientation of planar and linear structures, strike, dip, pitch, and plunge.			15
Texts / References	1. Holmes, A. (1992): Principles of Physical Geology, Vol.1, Chapman and Hall, London. 2. H. H. Read and J. Watson (1961).Introduction to Geology, Vol.1. Macmillan. 3. Grotzinger,J.,Jordan,T.H.,Press,F.,&Siever,R.(2010).Understanding Earth. Macmillan Higher Education. 4. Billings, M.P.(2016):Structural Geology, Pearson. 5. Park, R. G.(2017):Foundations of Structural Geology. Routledge. 6. Lisle, R.J.(2004):Geological Structures and Maps: A Practical Guide, Third edition. Elsevier			
Learning Outcomes (not more than 100 words)	The paper equips students with a fundamental understanding of Earth's physical attributes and its solar system context. Students will also gain insights into key geological theories and structural geology concepts.			

Course Title	ESMJ12: Practical connected with ESMJ11				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	---	01	01	
	Hour of Teaching (Total)	---	30	30	
Course Objectives (not more than 100 words)	The objective of this practical paper is to provide students with hands-on experience in geomorphologic models, the use of a clinometer compass, and exercises on structural problems, enhancing their practical skills in geological analysis and interpretation.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
	Study of important geomorphological models; study of clinometer compass; exercises on structural problems; completion of outcrops; drawing and interpretation of sections across elementary representative geological structures				30
Texts / References	1. Leet, L.D. and Judson, S.(1969): Physical Geology, Prentice Hall., Read, H.H. (1993) Introduction to Geology, 2. Billings, M.P.(2016):Structural Geology, Pearson. 3. Murck, B.W., & Skinner, B.J.(2015).Visualizing geology. John Wiley& Sons. 4. Lisle, R.J.(2004):Geological Structures and Maps: A Practical Guide, Third edition. Elsevier 5. Compton, R.A. (1985): Geology in the Field, John Wiley and Sons.				
Learning Outcomes (not more than 100 words)	Upon completing this paper, students will gain proficiency in analyzing important geomorphological models and using a clinometer compass. They will be adept at solving structural problems, completing outcrops, and drawing and interpreting sections across elementary representative geological structures.				

Course Title	ESSE11: Smartphone Geosciences				
Category of Course	SEC				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	02	---	02	
	Hour of Teaching (Total)	30	---	30	
Course Objectives (not more than 100 words)	To be effective, geoscience education is adapting to this changing environment by applying the use of smartphone technology. To implement experiential learning is considered instrumental to accomplishing the ready access to knowledge through Smartphone. Perceived ease of use of a smartphone in learning activities, The movability of smartphone devices has the potential to transform geosciences education, accessing worldwide knowledge system, and Entrepreneurship Skill Development.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Fundamentals of operating systems in smartphone, Geology Apps, AI Tools, ChatGPT, IoT, Multimedia Tools, Social media Apps for geology. MS Office Automation for report writing and presentation.				10
II	Smartphone assisted geological fieldwork, orientation of digital maps in field, marking location in digital map, bearing (front and back) using smartphone. GPS App, NAVIC, Digital Compass App.				10
III	Basic field measurement techniques using smartphone: Bedding dip and strike, Litholog measurement with smartphone, Development of digital catalogue with credentials of rock types, structures and texture of rocks, use of smartphone lens in field.				05
IV	Digital reading contours and topography using smartphone, Google Earth, Google Map, Bhuvan. Concepts of digital map reading, Distance, height, elevation and place approximation.				05
Texts / References	1. Elizabeth Woyke (2014) The Smartphone: Anatomy of an Industry. The New Press 2. Daniel Miller, et al., (2021) The Global Smartphone: Beyond a Youth Technology. UCL Press, pp 1-295. 3. Sujit Kumar Mishra (2020) Fundamentals of Android App Development. BPB Publications. 4. Francisco Yus (2022) Smart phone Communication: Interactions in the App Ecosystem. Routledge is an imprint of the Taylor & Francis Group, pp 1-318. 5. Michael J. Jipping (2020) Smartphone Operating System Concepts with Symbian OS: A Tutorial Guide. John Wiley & Sons, Ltd., England, pp 1-338. 6. Jeff McWherter and Scott Gowell (2012) Professional Mobile Application Development. John Wiley & Sons Inc. 7. Mike James (1984) Artificial Intelligence in BASIC. Newnes Technical Books, Butterworth & Co Publishers Ltd., pp 1-124. 8. Ahmed El- Rabbany (2007) Introduction to GPS: The Global Positioning System. Artech House, London, pp 1-176.				
Learning Outcomes (not more than 100 words)	Geoscientific exploration and classroom exercises can help diverse sets of students become more aware of the Earth and their environment by using basic apps on smart phones. Students will be able to achieve scientific literacy with digital content available in the public domains and internet portals. Adopt interactive learning through the use of quizzes, games, and flashcards over Smartphone. Manage a virtual classroom, send assignments, search and share useful materials, and communicate with students by Smartphone applications. Build awareness about ethical issues related to Social Media and Web Platforms.				

Course Title	ESSE12: Practical connected with ESSE11				
Category of Course	SEC				
Credits & Hour of Teaching		Theory	Practical	Cumulative	
	Credits	---	01	01	
	Hour of Teaching (Total)	---	30	30	
Course Objectives (not more than 100 words)	Application of smart phone is gradually becoming a compelling learning tool used to skill enhance education and entrepreneurship. To regulate practicality of smartphone in students' academic achievement with development of earning skill from digital resources.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Application digital apps using smartphone, measurements in open playground using digital GPS & NAVIC systems apps,				10
II	Geological Field Training with smartphone,				10
III	Multimedia Tools, Social media Apps for geology.				5
IV	Content Writing and Presentation using AI Tools.				5
Texts / References	1. Indian Space Research Organisation (ISRO): https://www.isro.gov.in. 2. ISRO-NRSC (2023) BHUVAN-SUVIDHA, DRISHTI: Manual for Field Data Collection using Smartphone application using Android. https://bhuvan.nrsc.gov.in. 3. National Aeronautics and Space Administration (NASA): https://www.nasa.gov. 4. Introduction: GooESe Earth User Guide. https://earth.gooESe.com. 5. Daniel Miller, et al., (2021) The Global Smartphone: Beyond a Youth Technology. UCL Press, pp. 1-295. 6. Peggy Anne Salz and Jennifer Moranz (2013) The Everything Guide to Mobile Apps: A Practical Guide to Affordable Mobile App Development for Your Business. PENGUIN. 7. Jerome DiMarzio and J DiMarzio (2016) Beginning Android Programming with Android Studio. Wrox Press. 8. Cameron Banga and Josh Weinhold (2014) <i>Essential Mobile Interaction Design: Perfecting Interface Design in Mobile Apps</i>. Addison-Wesley Professional.				
Learning Outcomes (not more than 100 words)	Students have learned the fundamentals of field techniques using smartphone and they have improved their field credentials with the application of smartphone apps in field trips and semester-long field courses, and summer field camps. Use smartphone technology in the classroom: managing groups, interacting, communicating, and sharing materials. Structure information, better analyzes, comprehend, synthesize, recall, and generate new ideas using technological tools.				

Course Title	ESMD11: Evolving Earth				
Category of Course	MD				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	03	---	03	
	Hour of Teaching (Total)	45	---	45	
Course Objectives (not more than 100 words)	This course aims to train the students: • about the basics of Geology, its scope and various branches of Geology with fundamental concepts of plate tectonics • To introduce the aspects of the Earth and Planetary system and its changes with time • To emphasize knowledge of the solar system, planets, and geological time scale. • To provide the fundamental ideas about different structures formed on earth surfaces.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Introduction to Earth Sciences, meaning and scope; Sub-disciplines and relationship with other branches of sciences				10
II	Plate tectonics; Geological time scale; Mass extinction; Ice ages				10
III	Introduction and origin of the solar system; Structure and composition of the earth.				10
IV	Basics of geological structures: bedding, fold and fault				15
Texts / References	1. Holmes, A. (1987): Principles of Physical Geology. Nelson, New York, 3rd edition. 2. Singh, S. (2003): Physical Geography (English and Hindi editions). PrayagPustakBhawan, Allahabad. 3. Hills, E. S. (1963): Elements of Structural Geology. Farrold and Sons, London. 4. Billings, M. P. (1972): Structural Geology. Prentice Hall.				
Learning Outcomes (not more than 100 words)	Upon completing this course, students will be able to: • Gain Knowledge about the history of Earth and planetary system and development with information on the formations of different types of plate boundaries • Acquire knowledge of the early solar system, Earth's structure and composition, and key events in past geological periods. • Know basic terminologies and concepts in structural geology				

Course Title	Same as major ESMJ11 and ESMJ12				
Category of Course	Minor				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	03	01	04	
	Hour of Teaching (Total)	45	30	75	

		Credit
Value Added Course(VA)	To be Offered by the University	2 + 2
Ability Enhancement Course(AE)	To be Offered by the University	2

Semester: II

Course Title	ESMJ21: Minerals and Earth Material				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	03	---	03	
	Hour of Teaching (Total)	45	---	45	
Course Objectives (not more than 100 words)	Identify the minerals and materials of the solid Earth. Recognize the most common rock-forming minerals, their compositions and to identify them based on their macroscopic and microscopic characteristics. To understand the fundamental principles of mineralogy, optical mineralogy and crystallography.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Fundamentals of Mineralogy: Concepts and classification of minerals; mineral forming processes (igneous or magmatism, sedimentation and metamorphism); General physical properties of minerals (color, form, habit, shape, luster, streak, cleavage, fracture, hardness, specific gravity) with other special properties (tenacity, odors/aromas/smells, transparency, hygroscopic, magnetic nature).				10
II	Mineralogy: Chemical composition and diagnostic physical properties of rock forming minerals mentioned below: olivine, pyroxene (augite, hypersthene), quartz, feldspar (orthoclase, microcline, albite, labradorite), nepheline, mica (muscovite, biotite), tremolite, hornblende, secondary minerals (serpentine, talc, chlorite, kaolinite), apatite, carbonate (calcite, dolomite), garnet, aluminosilicate (kyanite, sillimanite, andalusite), staurolite, topaz, tourmaline, corundum, gypsum, graphite.				10
III	Optical Mineralogy: Nature of light, optics of light, critical angle and total internal reflection, polarization, double refraction, Nicol prism, optically isotropic and anisotropic substances; sample preparation techniques for optical microscopy, polarizing microscope; common optical properties observed under plane polarized light (PPL) and crossed polarized light (CPL/XPL); optical properties of common rock-forming minerals: quartz, feldspar (orthoclase, microcline, plagioclase), garnet, mica (biotite, muscovite), pyroxene (augite, hypersthene), amphibole (hornblende), olivine, and calcite.				15
IV	Fundamentals of Crystallography: Crystal, elementary idea of crystal structure; Elements of crystal symmetry; Introduction to different crystals systems; Parts of crystal - face, edge, apex, solid angle and interfacial angle; Crystallographic axes and angles; Parameters and indices; Common crystal forms –Cube, rhombohedron, hexagons, octahedron, dodecahedron, dome, prism, pyramid and pinacoid;				10
Texts / References	- Berry, L.G., Mason, B. and Dietrich, R.V. (2004) Mineralogy. CBS Publication. - Ford, W.E. (2006) Danas Textbook of Mineralogy. CBS Publication. - Gribble, C.D. (2005) Rutley's Element of Mineralogy. CBS Publication. - Dexter Perkins (2015) Mineralogy. Pearson Education India. - Nesse, D.W. (2016) Optical Mineralogy, McGraw Hill Publication. - Kerr, P. F. (2014) Optical Mineralogy. McGraw-Hill. - Verma, P. K. (2010) Optical Mineralogy. Ane Books Pvt Ltd. - Phillips, F.C (1977) Introduction to Crystallography, Longman Group Publication. - Sharma, R.S. and Sharma, A. (2013) Crystallography and Mineralogy - Concepts and Methods. Geological society of India, Bengaluru. - Philpotts, A. and Klein, C. (2012) Earth Materials: Introduction to Mineralogy and Petrology. Cambridge University Press.				
Learning Outcomes (not more than 100 words)	The study of this course strengthens students' knowledge for understanding the fundamentals of the minerals and materials occur on the Earth. The knowledge of the concepts in physical properties of minerals would enable the students to understand the common natural resources of the earth and relate them to their field observations and also, in understanding the framework of the rock forming processes.				

Course Title		ESMJ22: Practical connected with ESMJ21			
Category of Course		Major			
Credits& Hour of Teaching			Theory	Practical	Cumulative
		Credits	---	01	01
		Hour of Teaching (Total)	---	30	30
Course Objectives (not more than 100 words)		Practical knowledge to recognize the most common rock-forming minerals, their compositions and to identify them based on their macroscopic and microscopic characteristics in hand specimen.			
Course Content		The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content				Hr. of Teaching
I	Learning of physical properties of minerals in hand specimen Silicates: Olivine, Garnet, Andalusite, Sillimanite, Kyanite, Staurolite, Beryl, Tourmaline, Augite, Actinolite, Tremolite, Hornblende, Serpentine, Talc, Muscovite, Biotite, Phlogopite, Quartz, Orthoclase, Plagioclase, Microcline, Nepheline, Sodalite, Zeolite.				10
II	Sulfides and Oxides of Copper, Sulfur, Graphite, Pyrite, Corundum, Magnetite Hydroxides, Halides, Carbonates, Sulfates, Phosphates: Psilomelane, Fluorite, Calcite, Malachite, Gypsum, Apatite.				10
III	Use of polarizing microscope; Study of optical properties of important rock forming minerals. Study of physical properties of minerals mentioned in theory course				5
IV	Study of elements of symmetry of representative crystals system: isometric, hexagonal, tetragonal, orthorhombic, monoclinic, triclinic				5
Texts / References		1. Gribble, C.D. (2005) Rutley’s Element of Mineralogy. CBS Publication. 2. Ford, W.E.(2006) Danas Text Book of Mineralogy, CBS Publication. 3. Dexter Perkins(2015) Mineralogy. Pearson Education India. 4. Nesse, D.W.(2016) Optical Mineralogy, McGraw Hill Publication. 5. Kerr, P.F.(2014)Optical Mineralogy. McGraw-Hill. 6. Verma, P.K.(2010)Optical Mineralogy. Ane Books PvtLtd. 7. Phillips, F.C (1977)Introduction to Crystallography, Longman Group Publication.			
Learning Outcomes (not more than 100 words)		The study of this course strengthens students’ knowledge forunderstanding the fundamentals of the minerals and materials occur on the Earth. The practical knowledge of the physical properties of minerals would enable the students to understand the common rock forming mineral on the Earth.			

Course Title	ESMD21: Earth, Environment and Society				
Category of Course	MD				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	03	---	03	
	Hour of Teaching (Total)	45	---	45	
Course Objectives (not more than 100 words)	This course aims to train the students • To associate the naturally occurring landforms with erosion and depositional action of the rivers, wind and Glaciers. • To know the brief ideas about concepts of hydrological cycles • Idea about the atmosphere and ocean system and its significance				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Physical division of India; Geological hazards: floods, earthquakes, volcanoes, tsunami, landslides. weathering and erosion, geological work of rivers, wind and Glaciers,				10
II	Atmosphere: structure and composition;				10
III	Hydrological cycle; groundwater/watershed management				10
IV	Introduction to ocean system. Significance,				10
V	Preservation and introduction of geo-heritage sites in India.				05
Texts / References	1. Gautam, A. (2006): Advanced Geography of India, ShardaPustakBhawan, Allahabad. 2. Bryant, E. (1985): Natural Hazards, Cambridge University Press. 3. Holmes, A. (1987): Principles of Physical Geology. Nelson, New York, 3rd edition. 4. Singh, S. (2003): Physical Geography (English and Hindi editions). PrayagPustakBhawan, Allahabad. 5. Lal, D. S. (2007): Climatology, ShardaPustakBhawan, Allahabad. 6. Strahler, A. N. and Strahler, A. H. (1973): Environmental Geoscience, Hamilton Publishing Co. 7. Davies, S. N. and De Wiest, R. J. N. (1966): Hydrogeology, John Wiley & Sons. 8. Garrison, T. (1998): Oceanography, Wordsworth Company, Belmont..				
Learning Outcomes (not more than 100 words)	Upon completing this course, students will be able to • Understand the depositional and erosional processes associated with various natural agents, such as Rivers, wind, glaciers, etc. • Acquire basic knowledge about the atmosphere, hydrological cycles and ocean system and their significance Know about Geo-heritage sites in India and their preservation				

Course Title	ESSE21: Gemology				
Category of Course	SEC				
Credits & Hour of Teaching		Theory	Practical	Cumulative	
	Credits	02	---	30	
	Hour of Teaching (Total)	30	---	30	
Course Objectives (not more than 100 words)	The course contains physical characterization and classification of gemstones, application of advanced techniques to gemstone identification, genesis of gemstones and Indian distribution, and Synthetic gemstones and their identification.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Gem and gemstones; General characteristics and chemical composition of gemstones; Physical characteristics - form, cleavage, fracture, hardness, specific gravity, and tenacity.				6
II	Optical characteristics - colour, luster, the play of colour, refractive index, reflectivity, diaphaneity pleochroism, dispersion; Classification of gemstones.				6
III	Application of ultraviolet rays, x-rays, and infra-red rays in gem identification; Electrical thermal and magnetic characters of gems;				6
IV	Systematic description, genesis, mode of occurrence, distribution in India, and important world occurrences of precious and semi-precious stones.				6
V	Synthetic gemstones - methods of synthesis and their characteristics and identification; Gem enhancement methods and their identification: Colourless / colored impregnation, heat treatment, coating, irradiation, diffusion, treatment.				6
Texts / References	1. Karanth, R. V. (2000): Gems and Gem Industry in India, Memoir-45, Geological Society of India, Bangalore. 2. Brocardo, G. (1981): Minerals and Gemstone - An Identification Guide, David and Charles, London. 3. Bruton, Eric F. G. A. (1970): Diamonds, Chilton Book Company. Max Bauer (1968): Precious Stones, Vol. I and II, Dover, New York. 4. Orlov, Yu. L. (1973): The Mineralogy of the Diamond, John Wiley. 5. Rajendran, S. (2007): Mineral Exploration: Recent Strategies. 6. Vilson, M. (1967): Gems, Heinemann, London.				
Learning Outcomes (not more than 100 words)	Gemology is a fascinating scientific field that entails the comprehensive study of both natural and artificial gemstone materials. As a specialized branch of mineralogy, gemology involves the meticulous examination of gemstones' physical and chemical properties to discern their unique characteristics. Further, it also helps to differentiate between natural and synthetic gemstones. This knowledge is decisive in ensuring the authenticity and value of gemstones in various applications, including jewelry, art, and scientific research.				

Course Title	ESSE22: Practical connected with ESSE21				
Category of Course	SEC				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	---	01	01	
	Hour of Teaching (Total)	---	30	30	
Course Objectives (not more than 100 words)	To identify the precious and semi-precious gems				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
	Study of diagnostic physical properties of precious and semi-precious gems.				30
Texts / References	1. Brocardo, G. (1981): Minerals and Gemstone - A identification guide, David and Charles, London. 2. Bruton Eric F. G. A. (1970): Diamonds, Chilton Book Company. 3. Max Bauer (1968): Precious Stones, Vol. I and II, Dover, New York.				
Learning Outcomes (not more than 100 words)	Students can able to identify gems in hand specimens using physical properties like hardness heat sensitivity, affect a gem's durability and cut and help gemologists with identifications.				

Course Title	Same as major ESMJ21 and ESMJ22				
Category of Course	Minor				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	03	01	04	
	Hour of Teaching (Total)	45	30	75	

		Credit
Value Added Course (VA)	To be Offered by the University	2 + 2
Ability Enhancement Course (AE)	To be Offered by the University	2

Semester: III

Course Title	ESMJ31: Physical Geology				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	03	---	03	
	Hour of Teaching (Total)	45	---	45	
Course Objectives (not more than 100 words)	Study the different geomorphic processes, Nature of weathering and erosion on the Earth's surface Evolution of landscape Concept of Isostasy and plate tectonics Erosional and depositional work of different natural agencies				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Exogenic and endogenic geomorphic processes; Weathering and erosion				10
II	Soil, soil formation, soil profile, and soil type;				5
III	Evolution of landscape; Origin of oceans, continents, and mountains; Classification of mountains;				10
IV	Concept and theories of isostasy; Origin and significance of mid-oceanic ridges and trenches; Seafloor spreading & continental drift; A brief idea about plate tectonics and distribution of plates.				10
V	A detailed account of the geological work of natural agencies; Glaciers, groundwater and springs, rivers, lakes, ocean, and wind.				10
Texts / References	1. Holmes, Arthur (1992): Principles of Physical Geology, Vol. 1, Chapman and Hall, London. 2. Leet, L. D. and Judson, S. (1969): Physical Geology, Prentice Hall. 3. Mallory, B. F. and Cargo, D. N. (1979): Physical Geology, McGraw Hill. 4. Monrow, James S. (1986): Physical Geology: Exploring the Earth, Booke Cole, Australia. 5. Ritter, Dale F. (1986): Processes of Geomorphology, Wm. C. Brown Publishers. 6. Singh, S. (2001): Geomorphology, PrayagPustakBhandar, Allahabad.				
Learning Outcomes (not more than 100 words)	This study covers both endogenic and exogenic geomorphic processes, which are important to understanding soil development and different landscapes. Isostasy and plate tectonics enhance our understanding of the Earth's dynamics. The erosion and depositional activity of different natural agencies and the development of related geomorphic features.				

Course Title	ESMJ32: Elementary Structural Geology			
Category of Course	Major			
Credits& Hour of Teaching		Theory	Practical	Cumulative
	Credits	03	---	03
	Hour of Teaching (Total)	45	---	45
Course Objectives (not more than 100 words)	The course aims to introduce undergraduate students to its basic principles, analysis methods and field characteristics. Overall goal is to provide a good understanding of how earth's rock deforms at different levels of the lithosphere. Students develop a basic appreciation of rock deformation at different temperatures and pressures and different scales and understand the difference between brittle and ductile deformation regimes. Students study diverse geometries and types of structures in the field and learn to relate them to the deformation regime.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	An introduction to various planar and linear geological structures, including the concepts of outcrop, and strike-dip. Detailed examination of the clinometer compass and its application in the field. Analysis of how topography influences outcrop patterns, with a focus on the application of rule of 'V's.			12
II	Definition and identification of the components of folds along with the geometrical classification of folds and methods for recognizing folds in the field. Discussion on the causes of folding and its geological significance.			12
III	Definition and classification of various types of unconformities. Introduction to the concepts of overlap, onlap, offlap, overstep, outlier, and inlier. Methods for recognizing unconformities in the field and discussion of their geological significance.			12
IV	Definition and detailed description of the components of faults and joints, with a focus on their geometric and genetic classification. Exploration of the effects of faulting on outcrops and the characteristic features that indicate faulting in the field. Examination of the causes and significance of faulting, with an emphasis on structures such as horst and graben, autochthon, allochthon, window, and klippe. Analysis of the characteristics of fractures and joints, including different fracture modes and their implications in geological studies.			9
Texts / References	1. Billings, M. P. (2016): Structural Geology, Pearson. 2. Dennis, J. G. (1972): Structural Geology, Ronald Press Company, New York. 3. Hills, E. S. (1963): Elements of Structural Geology, Farrold and Sons, London. 4. Park, R. G. (2017): Foundations of Structural Geology, Routledge. 5. Ramsay, J. G. (2004): Folding and Fracture of Rocks, The Blackburn Press. 6. Fossen, H. (2010): Structural Geology, Cambridge University Press.			
Learning Outcomes (not more than 100 words)	On completion of the course the student should have the following learning outcomes: Basic understanding of linear and planar structures and measuring, plotting and interpreting structural field data. Comprehend how to describe and classify brittle and ductile structures, including, folds, faults, joints and unconformities. Knows how to read simple geological maps and geological cross sections.			

Course Title	ESMJ33: Practical connected with ESMJ31				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	---	01	01	
	Hour of Teaching (Total)	---	30	30	
Course Objectives (not more than 100 words)	Practical exercises will be provided on what is learned about different types of physical structures in the Theory course (ES-CC-211).				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Study of Toposheets: Boundary Information of toposheet, Topographic features.				06
II	Cultural features: Man-made features, Land-use / Land cover.				06
III	Hydrological features: Drainage patterns.				06
IV	Extractions of Plateau, hill, Ridge, Valley, Escarpment, and Deep Gorges based on contours.				06
V	Study of Geomorphological features from Aerial photos under a pocket stereoscope.				06
Texts / References	1. Holmes, Arthur (1992): Principles of Physical Geology, Vol. 1, Chapman and Hall, London. 2. Leet, L. D. and Judson, S. (1969): Physical Geology, Prentice Hall. 3. Mallory, B. F. and Cargo, D. N. (1979): Physical Geology, McGraw Hill. 4. Monrow, James S. (1986): Physical Geology: Exploring the Earth, Booke Cole, Australia. 5. Ritter, Dale F. (1986): Processes of Geomorphology, Wm. C. Brown Publishers. 6. Singh, S. (2001): Geomorphology, PrayagPustakBhandar, Allahabad.				
Learning Outcomes (not more than 100 words)	After these practical exercises, the students will be able to visualize the physical features, geological structures, and man-made features on the toposheet. They will also be able to extract drainage patterns and different geomorphic features through Aerial photographs.				

Course Title	ESMJ34: Practical connected with ESMJ32				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	---	01	01	
	Hour of Teaching (Total)	---	30	30	
Course Objectives (not more than 100 words)	There will be practical exercises on what is learnt about different types of structures in the Theory course (ES-CC-212).				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Practice problems focused on determining true and apparent dips, strike, true and vertical thicknesses, and the width of outcrops in geological structures.				10
II	Calculation of strike, dip, and bed thicknesses using geological maps. Techniques for completing outcrop maps and the use of a clinometer compass for accurate measurement of dip and strike.				10
III	Interpretation of outcrops of folded rock units on both horizontal and sloping surfaces. Construction of geological cross-sections and analysis to infer the geological history of the region along with application of rule of 'V's.				10
Texts / References	1. Billings, M. P. (2016): Structural Geology, Pearson. 2. Lisle, R. J. (2004): Geological Structures and Maps: A Practical Guide, Third Edition. Elsevier. 3. Marshak, S. and Mitra, G. (1988): Basic Methods of Structural Geology, Prentice Hall. 4. Bennison, G. M. (1993): An Introduction to Geological Structures and Maps, Edward Arnold, London. 5. Ragan, D. M. (2009): Structural Geology: An Introduction to Geometrical Technique, Cambridge University Press. 6. Rowland, S. M., Duebendorfer, E. M., and Schiefelbein (2007): Structural Analysis & Synthesis: A Laboratory Course in Structural Geology, Blackwell Publishing Ltd.				
Learning Outcomes (not more than 100 words)	After going through these practical exercises, the students will be able to visualize the relationship between the physical features of the field and the geological structures. Student will also be able to make desired measurement of the subsurface structural features from the ground surface data.				

Course Title	ESMD31: Rocks, Minerals and Fossils				
Category of Course	MD				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	03	---	03	
	Hour of Teaching (Total)	45	---	45	
Course Objectives (not more than 100 words)	This course aims to train the students - to know the different types of rocks, such as Igneous, sedimentary, and metamorphic rocks and related textures, - to understand the ideas of understanding of different rock types based on their origin. - to train in identifying the properties of common rock-forming minerals and their occurrences in India To introduce fundamental aspects of paleontology and its application in Geology				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Rockcycle,abriefintroductionofIgneous,sedimentaryandmetamorphicrocks,				10
II	Physical and chemical properties of minerals, mineral resource of India.				15
III	Paleontology,definitionandscope,fossildefinition,modeofoccurrenceanduseoffossil.Geologicaltimescale				10
IV	Evolution of life is representative of life of different geological periods.				10
Texts / References	Cornelis Klein and A. R. Philpotts (2013):<i>Earth Materials: Introduction to Mineralogy and Petrology</i>. Cambridge University Press. Read, H. H. (1968):<i>Rutley's Element of Mineralogy</i> (Rev. Ed.). Thomas Murby and Co. Blatt, H. and Tracy, R. J. (1996):<i>Petrology: Igneous, Sedimentary, Metamorphic</i>. W. H. Freeman and Co., New York. Gokhale, K. V. G. K. and Rao, T. C. (1983):<i>Ore Deposits of India</i>. East-West Press Pvt. Ltd. Jain, P. C. and Anantharaman, M. S. (1983):<i>Palaeontology: Evolution and Animal Distribution</i>. Vishal Publ. Prothero, D. R. (2004):<i>Bringing Fossils to Life: An Introduction to Paleontology</i> (2nd Ed.). McGraw-Hill.				
Learning Outcomes (not more than 100 words)	Upon completing this course, students will be able to - Understand the different types of rock types and their formation in different tectonic environments - Know the fundamentals of mineralogy and common rock-forming minerals Basics of Paleontology and Distribution of different fossil assemblages in geological history.				

Course Title	ESSE31: Geological Field Training				
Category of Course	SEC				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	---	03	03	
	Hour of Teaching (Total)	---	90	90	
Course Objectives (not more than 100 words)	To impart the introductory level field training to the students to visualize various rock types in their natural environment and their natural relationship to one another. It seeks to describe and explain the surface feature and underground structure of the lithosphere based upon observations and inferences. To learn to use field equipments such as clinometers and other equipments and measuring and recording of field data and information. It seeks to describe.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	There will be a field visit in a suitable area with the following objectives: • To observe and identify the various ongoing weathering, erosional, and depositional features. • To acquaint with the various types of rocks observable in the area. • To distinguish between in situ and transported rocks. • To identify the various basic characters of the rocks on their outcrops. • Hands-on exercises with clinometer, compass, and other geological equipment. • Measuring and recording the dip and strikes of the rocks. • Marking the locations of the outcrops and plotting of the structural data on a base map. • To write a brief inventory report on the geological setup of the visited area.				90
Texts / References	1. Compton, R. A. (1985): Geology in the Field, John Wiley and Sons. 2. Freeman, T. (1999): Procedures in Field Geology, Blackwell Science. 3. Geological Field Techniques (2010), edited by Angela L. Coe, Wiley-Blackwell. 4. Lahee, F. H. (1987): Field Geology, Indian edn., CBS Publishers and Distributors.				
Learning Outcomes (not more than 100 words)	After going through these practical exercises in the field, the students will be able to visualize and experience the different geological processes that are going on the earth and inside the earth. Students will be able to know the occurrence of various rock types and different deformation structures found in the rocks in nature.				

Course Title	ESMV31: Minerals and Earth Material				
Category of Course	VOC				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	03	---	03	
	Hour of Teaching (Total)	45	---	45	
Course Objectives (not more than 100 words)	Identify the minerals and materials of the solid Earth. Recognize the most common rock-forming minerals, their compositions and be able to identify them based on their macroscopic and microscopic characteristics. To understand the fundamental principles of mineralogy, optical mineralogy and crystallography.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Fundamentals of Mineralogy: Minerals, definition and classification; Processes of mineral formation (magmatic, post-magmatic, pegmatitic, weathering, sedimentary, and metamorphic); Common physical properties of minerals (form and shape, color, streak, luster, cleavage, fracture, hardness, tenacity, transparency, specific gravity, magnetic nature).				10
II	Mineralogy: Chemical composition and diagnostic physical properties of rock-forming minerals mentioned below: quartz, orthoclase, microcline, albite, labradorite, nepheline, muscovite, biotite, augite, hypersthene, tremolite, hornblende, olivine, serpentine, talc, chlorite, apatite, calcite, dolomite, garnet, kyanite, sillimanite, andalusite, staurolite, topaz, tourmaline, corundum, gypsum, graphite, and kaolinite.				10
III	Optical Mineralogy: Nature of Light, Optics of Light, Critical Angle and Total Internal Reflection, Polarization of Light, Double Refraction, Nicol Prism and Its Construction, Sample Preparation Techniques for Optical Microscopy, Polarizing Microscope, Its Parts and Functioning; Optically Isotropic and Anisotropic Substances, Common Optical Properties Observed Under Polarized Light and Crossed Nicols; Optical Properties of Some Common Rock-Forming Minerals (quartz, orthoclase, microcline, plagioclase, garnet, biotite, muscovite, augite, hypersthene, hornblende, olivine, and calcite).				15
IV	Fundamentals of Crystallography: Crystal, elementary idea of crystal structure; Parts of crystal - face, edge, apex, solid angle and interfacial angle; Crystallographic axes and angles; Parameters and indices; Common crystal forms - dome, prism, pyramid, and pinacoid; Elements of crystal symmetry; Introduction to different crystal systems.				10
Texts / References	1. Berry, L. G., Mason, B., and Dietrich, R. V. (2004): Mineralogy, CBS Publication. 2. Ford, W. E. (2002): Dana's Textbook of Mineralogy, CBS Publication. 3. Gribble, C. D. (2005): Rutley's Elements of Mineralogy, CBS Publication. 4. Perkins, Dexter (2015): Mineralogy, Pearson Education India. 5. Nesse, D. W. (2016): Optical Mineralogy, McGraw-Hill Publication. 6. Kerr, P. F. (2014): Optical Mineralogy, McGraw-Hill. 7. Verma, P. K. (2010): Optical Mineralogy, Ane Books Pvt Ltd. 8. Phillips, F. C. (1977): Introduction to Crystallography, Longman Group Publication. 9. Sharma, R. S. and Sharma, A. (2013): Crystallography and Mineralogy - Concepts and Methods, Geological Society of India, Bengaluru. 10. Philpotts, A. and Klein, C. (2012): Earth Materials: Introduction to Mineralogy and Petrology, Cambridge University Press.				
Learning Outcomes (not more than 100 words)	The study of this course strengthens students' knowledge for understanding the fundamentals of the minerals and materials occur on the Earth. The knowledge of the concepts in physical properties of minerals would enable the students to understand common natural resources of the earth and relate them to their field observations and also, in understanding the framework of the rock forming processes.				

Course Title	ESMV32: Practical connected with ESMV31				
Category of Course	Minor				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits		01	01	
	Hour of Teaching (Total)		30	30	
Course Objectives (not more than 100 words)	Practical knowledge to recognize the most common rock-forming minerals, their compositions and be able to identify them based on their macroscopic and microscopic characteristics in hand specimens.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Study of physical properties of minerals in hand specimen Silicates: Olivine, Garnet, Andalusite, Sillimanite, Kyanite, Staurolite, Beryl, Tourmaline, Augite, Actinolite, Tremolite, Hornblende, Serpentine, Talc, Muscovite, Biotite, Phlogopite, Quartz, Orthoclase, Plagioclase, Microcline, Nepheline, Sodalite, Zeolite.				10
II	Sulfides and Oxides of Copper, Sulfur, Graphite, Pyrite, Corundum, Magnetite, Hydroxides, Halides, Carbonates, Sulfates, Phosphates: Psilomelane, Fluorite, Calcite, Malachite, Gypsum, Apatite.				10
III	Use of polarizing microscope; Study of optical properties of important rock forming minerals. Study of physical properties of minerals mentioned in theory course.				5
IV	Study of elements of symmetry of representative crystals from each system mentioned in theory course.				5
Texts / References	1. Gribble, C. D. (2005): Rutley's Elements of Mineralogy, CBS Publication. 2. Ford, W. E. (2006): Dana's Textbook of Mineralogy, CBS Publication. 3. Perkins, Dexter (2015): Mineralogy, Pearson Education India. 4. Nesse, D. W. (2016): Optical Mineralogy, McGraw-Hill Publication. 5. Kerr, P. F. (2014): Optical Mineralogy, McGraw-Hill. 6. Verma, P. K. (2010): Optical Mineralogy, Ane Books Pvt Ltd. 7. Phillips, F. C. (1977): Introduction to Crystallography, Longman Group Publication.				
Learning Outcomes (not more than 100 words)	The study of this course strengthens students' knowledge for understanding the fundamentals of the minerals and materials occur on the Earth. The practical knowledge of the physical properties of minerals would enable the students to understand common rock forming mineral on the Earth.				

		Credit
Ability Enhancement Course (AE)	To be Offered by the University	2

Semester: IV

Course Title	ESMJ41: Petrology and Economic Geology				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	03	---	03	
	Hour of Teaching (Total)	45	---	45	
Course Objectives (not more than 100 words)	• To understand the origin, composition, and evolution of igneous rocks, and the processes of formation, including melting, crystallization, magmatic differentiation, and assimilation. • To focus on identifying common sedimentary rocks, the classification and the petrographic description and sedimentary structures. • To recognize metamorphic rocks, textures, structures, and assemblages and to provide details of metamorphic processes, agents, and types of metamorphism. • To have a basic idea about concepts of economic geology, and economic resources of India such as minerals, rocks, coal and petroleum.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Igneous Petrology: Magma: definition, its composition and types; Phase rule and its application to H_2O and SiO_2 systems; Common igneous textures; Magmatic differentiation and assimilation; Introduction to mineralogical classification; Brief petrographic description of common igneous rocks (granite, diorite, syenite, gabbro, dolerite, basalt, rhyolite, trachyte, pyroxenite and peridotite)				11
II	Sedimentary Petrology: Sedimentary rocks and their significance; Formation of sedimentary rocks; Classification of sedimentary rocks; Primary sedimentary structures (bedding and stratification, ripple marks, cross-bedding, graded bedding, mud cracks, raindrop imprints); Texture of sedimentary rocks; Sedimentary rock-forming minerals; Petrographic description of important rock types (sandstone, siltstone, conglomerate, breccia, shale, limestone, dolostone).				11
III	Metamorphic Petrology: Introduction, Limits of metamorphism, Factor controlling metamorphism, Types of metamorphism, Nomenclature and classification of metamorphic rocks, Recognition of rock-structure and textures, Metamorphic grade and isogrades, Regional, thermal, and progressive metamorphism of mafic igneous, calcareous, and pelitic rocks, Common metamorphic rocks such as slate, phyllite, schist, gneiss, hornfels, marble, and quartzite, along with their parent compositions (protoliths).				11
IV	Economic Geology: Definitions of Economic Geology, Ore Geology, Mineral Deposit, Ore Deposit, Ore, Ore Mineral, Gangue, Grade, Resource, and Reserve; Classification of Mineral Deposits; Styles of Mineralization and Morphology of Mineral Deposits; Chemical composition, diagnostic characters, uses, and Indian distribution of: Magnetite, Hematite, Chromite, Psilomelane, Pyrolusite, Chalcopyrite, Galena, Sphalerite, Gold, Magnesite, Bauxite, Pyrite, Diamond, Muscovite, Beryl, Fluorite, Gypsum, Barite, Halite, Phosphorite, Talc, Kyanite, Graphite, Asbestos, Monazite, and Corundum; Definition of Coal; Physical characteristics and composition of Coal; Classification of Coal; Uses of Coal; Distribution of Coal Deposits in India; Elementary idea about Petroleum, Uses, and distribution in India..				12

Texts / References	1. Best, M. G. (2002).<i>Igneous and Metamorphic Petrology</i>. Blackwell Science. 2. Blatt, H., & Tracy, R. J. (1996).<i>Petrology: Igneous, Sedimentary, Metamorphic</i>. W. H. Freeman and Co., New York. 3. Ehlers, E. G., & Blatt, H. (1982).<i>Igneous, Sedimentary and Metamorphic Petrology</i>. CBS Publ. Huang. 4. Winkler, H. G. F. (1967).<i>Petrogenesis of Metamorphic Rocks</i>. Springer-Verlag. 5. Winter, J. D. (2010).<i>Principles of Igneous and Metamorphic Petrology</i> (2nd ed.). Pearson Education Ltd. 6. Tucker, M. E. (2001).<i>Sedimentary Petrology: An Introduction to the Origin of Sedimentary Rocks</i>. 7. Sengupta, S. (1994).<i>Introduction to Sedimentology</i>. Routledge. 8. Boggs, S. (2013).<i>Principles of Sedimentology and Stratigraphy</i>. Pearson. 9. Bateman, A. M., & Jensen, M. L. (1990).<i>Economic Mineral Deposits</i>. John Wiley. 10. Chandra, D., & Singh, R. M. (2003).<i>Petroleum (Indian Context)</i>. Tara Book Agency. 11. Chandra, D., Singh, R. M., & Singh, M. P. (2000).<i>Textbook of Coal (Indian Context)</i>. Tara Book Agency. 12. Evans, A. M. (1993).<i>Ore Geology and Industrial Minerals: An Introduction</i>. John Wiley & Sons. 13. Evans, A. M. (1997).<i>An Introduction to Economic Geology and its Environmental Impact</i>. John Wiley & Sons. 14. Misra, K. (2000).<i>Understanding Mineral Deposits</i>. Springer Science & Business Media. 15. Pohl, W. L. (2011).<i>Economic Geology: Principles and Practice</i>. John Wiley & Sons. 16. Prasad, U. (2006).<i>Economic Geology: Economic Mineral Deposits</i>. CBS Publishers. 17. Tyrrell, G. W. (1978).<i>The Principles of Petrology</i>. Springer. 18. Klein, C. (2008).<i>Minerals and Rocks: Exercises in Crystal and Mineral Chemistry, Crystallography, X-Ray Powder Diffraction, Mineral and Rock Identification, and Ore Mineralogy</i>. John Wiley & Sons. 19. Perkins, D. (2014).<i>Mineralogy</i>. Pearson Education Limited. 20. Verma, P. K. (2010).<i>Optical Mineralogy (Four Colour)</i>. Ane Books Pvt Ltd.
Learning Outcomes (not more than 100 words)	The students are expected to have basic knowledge of • Igneous rocks mineral composition, texture, and other physical characteristics. Processes of magma generation, differentiation, and crystallization, and how these processes influence the composition and texture of igneous rocks. • Sedimentary rocks, the basic process of sedimentation (physical and chemical), deposition and diagenesis. Different classification of sedimentary rocks, their structures, textures and variability. • The metamorphism processes and metamorphic textures. Understand the conditions of metamorphism and the protolith of metamorphic rocks. • Basic concepts of economic geology and economic significance of certain mineral resources. Elementary knowledge of coal and petroleum.

Course Title	ESMJ42: Paleontology and Stratigraphy				
Category of Course	Major				
Credits & Hour of Teaching		Theory	Practical	Cumulative	
	Credits	03	---	03	
	Hour of Teaching (Total)	45	---	45	
Course Objectives (not more than 100 words)	To study the modes of preservasions of fossils and their morphology which is crucial for understanding the overall history of life and earth?However, stratigraphy aims at the organization of rock sequences in their proper order of occurrence according to their relative time of formation.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Paleontology:Definition, subdivision and scope of paleontology; Relationship of paleontology with other branches of geology; Fossils, their different kinds,requisites and various modes of preservation; Incompleteness of fossil records; Ideas about the evolution of life and adaptation to different environments.				10
II	Systematic classification of organisms; Study of the important morphological characters and geological distribution of Trilobita, Graptoloidea, Anthozoa and Echinoidea.				14
III	Stratigraphy:Definition and scope of stratigraphy, its relationship with other branches of geology; Laws of stratigraphy;Geological Time Scale; Stratigraphic nomenclature and classification; rock units, time units and time-rock units; Physica land tectonic divisions of India and their salient features.				07
IV	Concept of craton; Mobile belt and shield; Tectonic framework of the Indian shield; Evolution of granite-greenstone belts; Indian cratons (Aravalli, Dharwar, Singhbhum, Bastar and Bundelkhand); Eastern Ghats mobilebelt, Southern Granulite Terrain and the Central IndianTectonic Zone;Precambrians of the Extra-peninsular India.				14
Texts / References	1. Black, R. M. (1988).<i>The Elements of Palaeontology</i>. Cambridge Univ. 2. Boggs, S. (2001).<i>Principles of Sedimentology and Stratigraphy</i>. Prentice Hall. 3. Krishnan, M. S. (1968).<i>Geology of India and Burma</i>. Higgibothon, Madras. 4. Kumar, R. (1985).<i>Historical Geology and Stratigraphy of India</i>. Wiley Eastern Ltd. 5. Benton, Michael J., & Harper, David A. T. (2020).<i>Introduction to Paleobiology and the Fossil Record</i>. John Wiley & Sons Ltd. 6. Foote, M., & Miller, A. I. (2007).<i>Principles of Paleontology</i>. John Wiley & Sons Ltd. 7. Brookfield, M. E. (2004).<i>Principles of Stratigraphy</i>. Wiley Blackwell. 8. Moore, R. C., Lalicker, C. G., & Fischer, A. G. (1997).<i>Invertebrate Fossils</i>. CBS Publ. 9. Nield, E. W., & Tucker, V. C. T. (1985).<i>Palaeontology: An Introduction</i>.Pergamon Press. 10. Prothero, D. R. (2004).<i>Bringing Fossils to Life: An Introduction to Paleontology</i> (2nd ed.). McGraw-Hill. 11. Shrock, R. R., &Twenhoffel, W. H. (1952).<i>Principles of Invertebrate Paleontology</i>. CBS Publ. 12. Wadia, D. N. (1966).<i>Geology of India</i>.English Language Publ. 13. Weller, J. M. (1960).<i>Stratigraphic Principles and Practices</i>. Universal Book. 14. Woods, H. (1985).<i>Palaeontology Invertebrate</i>. CBS Publ. 15. Jain, P. C., &Anantharaman, M. S. (1983).<i>Palaeontology: Evolution and Animal Distribution</i>. Vishal Publ.				
Learning Outcomes (not more than 100 words)	By the end of this course, students will understand the scope of paleontology and its relationship with other geological fields. They will be proficient in identifying and classifying fossils, understanding fossilization processes, and the origin of life. Students will analyze the morphology and distribution of various fossil groups. In stratigraphy, they will grasp its principles, the Geological Time Scale, stratigraphic classification, and the structural subdivisions of India. Additionally, students will comprehend the tectonic framework of the Indian shield, including cratons, mobile belts, and Precambrian of extra-peninsular regions.This will provide a holistic idea of Indian geology to students.				

Course Title	ESMJ43:Environmental Geology and Hydrogeology				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	03	---	03	
	Hour of Teaching (Total)	45	---	45	
Course Objectives (not more than 100 words)	This course aims to comprehensively understand Earth's processes and their interaction with human activities. Key objectives include examining geological hazards such as landslides and earthquakes and their societal impact. It also focuses on groundwater systems, contaminant transport, and remediation strategies to safeguard water resources. The course emphasizes practical applications in environmental management and sustainability, equipping students with skills to assess geological factors in land-use planning, infrastructure development, and natural resource conservation. Through case studies and fieldwork, students develop critical thinking and problem-solving abilities essential for addressing contemporary environmental challenges				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Environmental Geology: Introduction to Environmental Geology and its domains. Population, Growth & Sustainability; environmental risk, human interaction & conflict.				10
II	Weather & Climate: Climate classification and change. Atmospheric composition and hazards; Ozone depletion, Lithosphere—Earthquakes & seismic hazard, Landslides. Human impact on environment; Environmental legislation.				10
III	Hydrogeology: Definition of hydrogeology, hydrological cycle and the role of groundwater in the hydrological cycle, hydrological components: precipitation, infiltration, evaporation & transpiration, origin of groundwater, vertical distribution of groundwater, types of aquifers, springs, hydrogeological properties of water-bearing rock—porosity, permeability, specific yield and specific retention, storage coefficient, hydraulic conductivity.				15
IV	Groundwater Quality: Physical and chemical properties of water, quality criteria for different uses. Introduction to surface and subsurface methods of groundwater exploration.				10
Texts / References	Environmental Geology 1. Reichard, J. S. (2018).<i>Environmental Geology</i>. McGraw-Hill. 2. Valdiya, K. S. (2013).<i>Environmental Geology</i>. McGraw-Hill. 3. Montgomery, C. W., & Szablewski, G. S. (2023).<i>Environmental Geology</i>. McGraw-Hill. 4. Keller, E. A. (2012).<i>Introduction to Environmental Geology</i>. Prentice Hall. 5. Erikson, J. (2002).<i>Environmental Geology: Facing the Challenges on Our Changing Earth</i>. Facts on File. Hydrogeology 1. Todd, D. K. (2006).<i>Groundwater Hydrology</i> (2nd ed.). John Wiley & Sons, N.Y. 2. Karanth, K. R. (1989).<i>Hydrogeology</i>. Tata McGraw-Hill Publishing. 3. Raghunath, H. M. (1990).<i>Groundwater</i>. Wiley Eastern Ltd. 4. Subramaniam, V. (2000).<i>Water</i>. Kingston Publishing, London.				
Learning Outcomes (not more than 100 words)	Upon completing Environmental Geology and Hydrogeology, students will: 1. Understand geological processes affecting environmental quality, including hazards like landslides and earthquakes. 2. Analyze hydrogeological systems, including groundwater flow, contamination mechanisms, and remediation strategies. 3. Apply geological knowledge to assess the environmental impacts of human activities on landscapes and water resources. 4. Develop mapping and interpreting geological data skills to support environmental planning and resource management. 5. Evaluate sustainable practices and policies for mitigating geological risks and preserving natural environments. Demonstrate proficiency in communicating geological concepts and solutions to diverse stakeholders, promoting informed decision-making in environmental management.				

Course Title		ESMJ44:Essentials of Remote Sensing			
Category of Course		Major			
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	02	---	02	
	Hour of Teaching (Total)	30	---	30	
Course Objectives (not more than 100 words)		The course aims to develop a basic understanding of Remote sensing and different space missions. It also includes the different image interpretation techniques and identification of landforms through remote sensing data.			
Course Content		The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content				Hr. of Teaching
I	Basic concepts of remote sensing; Indian space mission.				06
II	Elementsofphotogeology;Aerialphotos–types, scale, resolution, properties of aerial photos, stereoscopic parallax, relief displacement, vertical exaggeration.				06
III	Types of Remote Sensing Data; Resolution				06
IV	Elements of Image Interpretation;				06
V	Elements of Image Patterns–Landforms, Drainage, Erosion Details,etc.				06
Texts / References	1. Gupta, R. P. (1991).<i>Remote Sensing Geology</i>. Springer-Verlag. 2. Lillesand, T. M., & Kiefer, R. W. (1987).<i>Remote Sensing and Image Interpretation</i>. 3. Siegal, B. S., & Gillespie, A. R. (1980).<i>Remote Sensing in Geology</i>. John Wiley. 4. Campbell, J. B. (2002).<i>Introduction to Remote Sensing</i>(5th ed.). Taylor & Francis, London. 5. Allum, J. A. E. (1985).<i>Photogeology and Regional Mapping</i>. 6. Curran, P. J. (1985).<i>Principles of Remote Sensing</i>. Longman, London. 7. Deekshatulu, B. L., &Rajan, Y. S. (Eds.) (1984).<i>Remote Sensing</i>.Indian Academy of Science, Bangalore. 8. Pandey, S. N. (2007).<i>Principles and Applications of Photogeology</i>.				
Learning Outcomes (not more than 100 words)	Upon completion of this course, students will have a thorough understanding of the physics of geospatial technology, as well as a better understanding of the interpretation of remote sensing data.				

Course Title	ESMJ45: Practical connected with ESMJ41				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	---	01	01	
	Hour of Teaching (Total)	---	30	30	
Course Objectives (not more than 100 words)	Megascopic and microscopic study of the following rock types: gneiss, schist, quartzite, marble, phyllite and slate. To observe the physical and optical properties of certain minerals of economic importance in hand specimens and to know the Indian distribution of major mineral deposits, coal deposits and petroleum deposits by plotting their location of major deposits on the map of India.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Petrology: Megascopic and microscopic study of the following rock types: granite, syenite, nephelinesyenite, diorite, gabbro, peridotite, rhyolite, trachyte, dolerite, basalt, sandstone, conglomerate, breccia, shale, limestone, gneiss, schist, quartzite, marble, phyllite, and slate..				15
II	Economic Geology: Study of ore and economic minerals in hand specimens as detailed in the theory syllabus; Preparation of maps showing the distribution of important metallic and non-metallic mineral deposits, and important coal and oilfields of India.				15
Texts / References	1. MacKenzie, W. S., Donaldson, C. H., and Guilford, C., (1982). Atlas of Igneous Rocks and Their Textures. 2. Lewis D.W., and McConchie D., (2012). Practical Sedimentology. 3. Prasad, U., (2019). Economic Geology CBS Publishers & Distributors, New Delhi. 4. Deb, M., and Sarkar S. C., (2017) Minerals and Allied Natural Resources and their Sustainable Development. Springer Singapore. 5. Adams, A. E., MacKenzie, W. S., and C. Guilford., (1984). Atlas of Sedimentary Rocks Under the Microscope. 6. Yardley, B., W. D., MacKenzie, W. S., and Guilford, C., (1990) Atlas of Metamorphic Rocks and Their Textures				
Learning Outcomes (not more than 100 words)	In practical of igneous, sedimentary, metamorphic petrology and economic geology, students will gain the ability to identify and classify various rock types and their textures using hand samples and thin sections. They will develop skills in interpreting magmatic processes, depositional environments, and metamorphic conditions through physical and optical properties. Additionally, they will enhance their understanding of resource exploration by evaluating economic minerals and deposits and integrating practical data with theoretical concepts.				

Course Title	ESMJ46: Practical connected with ESMJ42			
Category of Course	Major			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	---	01	01
	Hour of Teaching (Total)	---	30	30
Course Objectives (not more than 100 words)	There will be practical exercises to learn about different types of fossils and rock types and their modes of occurrence in India.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Palaeontology: Study of modes of preservation of fossils.			5
II	Study of morphological characters of about 30 genera pertaining to Trilobita, Graptoloidea, Echinoidea, and Anthozoa..			10
III	Stratigraphy: Preparation of lithostratigraphic maps of India showing distribution of the following –Aravalli, Dharwar, and Singhbhum Supergroups.			7
IV	Study of important rock types of the above mentioned stratigraphic units			8
Texts / References	1. Black, R. M. (1988). <i>The Elements of Palaeontology</i>. Cambridge Univ. 2. Boggs, S. (2001). <i>Principles of Sedimentology and Stratigraphy</i>. Prentice Hall. 3. Krishnan, M. S. (1968). <i>Geology of India and Burma</i>. Higgibothon, Madras. 4. Kumar, R. (1985). <i>Historical Geology and Stratigraphy of India</i>. Wiley Eastern Ltd. 5. Benton, Michael J., & Harper, David A. T. (2020). <i>Introduction to Paleobiology and the Fossil Record</i>. John Wiley & Sons Ltd. 6. Foote, Michael, & Miller, Arnold I. <i>Principles of Paleontology</i>. John Wiley & Sons Ltd. 7. Brookfield, Michael E. <i>Principles of Stratigraphy</i>. Wiley Blackwell. 8. Moore, R. C., Lalicker, C. G., & Fischer, A. G. (1997). <i>Invertebrate Fossils</i>. CBS Publ. 9. Nield, E. W., & Tucker, V. C. T. (1985). <i>Palaeontology: An Introduction</i>. Pergamon Press. 10. Prothero, D. R. (2004). <i>Bringing Fossils to Life: An Introduction to Paleontology</i> (2nd ed.). McGraw-Hill. 11. Shrock, R. R., & Twenhoffel, W. H. (1952). <i>Principles of Invertebrate Paleontology</i>. CBS Publ. 12. Wadia, D. N. (1966). <i>Geology of India</i>. English Language Publ. 13. Weller, J. M. (1960). <i>Stratigraphic Principles and Practices</i>. Universal Book. 14. Woods, H. (1985). <i>Palaeontology Invertebrate</i>. CBS Publ. 15. Jain, P. C., & Anantharaman, M. S. (1983). <i>Palaeontology: Evolution and Animal Distribution</i>. Vishal Publ.			
Learning Outcomes (not more than 100 words)	After going through these practical exercises, the students will be able to identify different types of fossils and their modes of preservation in nature. Students will also be able to identify different types of stratigraphic rocks and their geological importance.			

Course Title	ESMJ47: Practical connected with ESMJ43			
Category of Course	Major			
Credits& Hour of Teaching		Theory	Practical	Cumulative
	Credits	---	01	01
	Hour of Teaching (Total)	---	30	30
Course Objectives (not more than 100 words)	The Environmental Geology and Hydrogeology theory course aims to provide students with a comprehensive understanding of the Earth's geological processes and their impact on water resources. Objectives include developing knowledge of soil and rock formation, groundwater flow dynamics, and the interactions between geological formations and water systems; understanding the principles of aquifer properties and groundwater contamination; learning to assess the environmental impact of geological and human activities on water resources; and equipping students with the skills to apply geological and hydrogeological concepts in environmental assessments, resource management, and sustainability practices. This course prepares students for advanced studies and careers in geosciences and environmental management.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Groundwater table contouring from piezo meter data, demarcation of flow direction,statusofsurface and subsurface water interaction on water table contour maps;Physico-chemical analysis ofgroundwater; Ion balancing of water chemistry data to estimate error in analysis, and methods ofgraphical representation of water quality data and its application, Earthquake hazard zonation;Landslidehazardzonation.			30
Texts / References	Environmental Geology 1. Reichard, J. S. (2018).<i>Environmental Geology</i>. McGraw-Hill. 2. Valdiya, K. S. (2013).<i>Environmental Geology</i>. McGraw-Hill. 3. Montgomery, C. W., &Szablewski, G. S. (2023).<i>Environmental Geology</i>. McGraw-Hill. 4. Keller, E. A. (2012).<i>Introduction to Environmental Geology</i>. Prentice Hall. 5. Erikson, J. (2002).<i>Environmental Geology: Facing the Challenges on Our Changing Earth</i>. Facts on File. Hydrogeology 1. Todd, D. K. (2006).<i>Groundwater Hydrology</i> (2nd ed.). John Wiley & Sons, N.Y. 2. Karanth, K. R. (1989).<i>Hydrogeology</i>. Tata McGraw-Hill Publishing. 3. Raghunath, H. M. (1990).<i>Groundwater</i>. Wiley Eastern Ltd. 4. Subramaniam, V. (2000).<i>Water</i>. Kingston Publishing, London.			
Learning Outcomes (not more than 100 words)	Upon completing the Environmental Geology and Hydrogeology Practical course, students will: 1. Gain hands-on experience identifying and analyzing geological materials and groundwater systems. 2. Learn to conduct field sampling, perform laboratory tests on soil and rock properties, and analyze groundwater flow and contamination using advanced equipment. 3. Ability to interpret geological maps, applies hydrogeological modeling techniques, and assesses the impact of human activities on water resources. 4. Develop data analysis, critical thinking, and problem-solving skills, preparing them to tackle environmental and hydrological issues effectively. Essential for environmental consulting, research, and resource management careers.			

Course Title	Same as major ESMJ43and ESMJ47			
Category of Course	Minor			
Credits& Hour of Teaching		Theory	Practical	Cumulative
	Credits	03	01	04
	Hour of Teaching (Total)	45	30	75
				Credit
Ability Enhancement Course (AE)		To be Offered by the University		2

Semester: V

Course Title	ESMJ51: Metamorphic Petrology				
Category of Course	Major				
Credits & Hour of Teaching		Theory	Practical	Cumulative	
	Credits	03	---	03	
	Hour of Teaching (Total)	45	---	45	
Course Objectives (not more than 100 words)	The course aims to provide a comprehensive understanding of the principles governing metamorphic processes, including phase rule application and mineralogical phase rules. Students will learn to classify metamorphic facies and facies series, analyze mineral assemblages using ACF, AKF, and AFM diagrams, and explore the progressive metamorphism of different rock types. Additionally, the course will cover metamorphic differentiation, the impact of plate tectonics on metamorphism, and the application of thermodynamic laws to metamorphic petrology.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Introduction, Phase Rules, Gibbs phase Rule, Goldschmidt's Mineralogical Phase Rule, Principles of metamorphic reactions, Classification and concepts of Metamorphic Facies and Facies Series				09
II	Phase diagram for metamorphic mineral assemblages, ACF, AKF, AFM diagrams, Prograde, retrograde and polymetamorphism.				09
III	Progressive metamorphism of (a) Ultramafic rocks in MgO – Al ₂ O ₃ – SiO ₂ – H ₂ O diagram, (b) Basic rocks in CaO – FeO – MgO – Al ₂ O ₃ – SiO ₂ diagrams, (c) Pelitic rocks in K ₂ O – FeO-MgO-Al ₂ O ₃ -SiO ₂ diagram, (d) Calcareous rocks in CaO – MgO –SiO ₂ – CO ₂ – H ₂ O diagram.				09
IV	Relationship between metamorphism and plate tectonics, Metamorphic Differentiation				09
V	Thermodynamic laws and their applications in metamorphic petrology				09
Texts / References	1. Blatt, H., & Tracy, R. J. (1996).<i>Petrology: Igneous, Sedimentary, Metamorphic</i>, W. H. Freeman & Co., New York. 2. Bucher, K., & Martin, F. (2002).<i>Petrogenesis of Metamorphic Rocks</i> (7th Rev. ed.). Springer-Verlag. 3. Ehlers, E. G., & Blatt, H. (1982).<i>Igneous, Sedimentary and Metamorphic Petrology</i>. CBS Publ. 4. Mason, R. (1978).<i>Petrology of Metamorphic Rocks</i>. CBS Publ. 5. Philpotts, A. R. (1994).<i>Principles of Igneous and Metamorphic Petrology</i>. Prentice Hall. 6. Powell, R. (1978).<i>Equilibrium Thermodynamics in Petrology: An Introduction</i>. Harper & Row Publishers, London. 7. Rastog, R. P., & Mishra, R. R. (1993).<i>An Introduction to Chemical Thermodynamics</i>. Vikash Publishing House. 8. Spry, A. (1976).<i>Metamorphic Textures</i>. Pergamon Press. 9. Vernon, R. H., & Clarke, G. L. (2008).<i>Principles of Metamorphic Petrology</i>. Cambridge University Press. Cambridge, New York, Melbourne, Madrid, Cape Town, Singapore, São Paulo, Delhi. 10. Winkler, H. G. C. (1967).<i>Petrogenesis of Metamorphic Rocks</i>. Narosa Publ. 11. Winter, J. D. (2001).<i>An Introduction to Igneous and Metamorphic Petrology</i>. Prentice Hall. 12. Wood, B. J., & Fraser, D. G. (1976).<i>Elementary Thermodynamics for Geologists</i>. Oxford University Press, London. 13. Yardley, B. W. D. (1989).<i>An Introduction to Metamorphic Petrology</i>. Longman Scientific and Technical, New York.				
Learning Outcomes (not more than 100 words)	The course will assist to understand the metamorphic processes in depth knowledge. Students will learn to classify metamorphic facies and chemographic diagrams for mineral assemblage projections of different metamorphic rocks. Students will learn the importance of metamorphism to reconstruct the tectonic plates.				

Course Title	ESMJ52: Sedimentary Petrology				
Category of Course	Major				
Credits & Hour of Teaching		Theory	Practical	Cumulative	
	Credits	03	---	03	
	Hour of Teaching (Total)	45	---	45	
Course Objectives (not more than 100 words)	This course is designed to give knowledge related to sedimentary processes such as weathering erosion and sedimentation. Also, the students will learn about the classification of various sedimentary rock types, their textures, mineralogy and diagenetic realms. They will get acquainted with various sedimentary structures and petrographic details of various sedimentary rock. Furthermore, this course will impart knowledge about various depositional environments and provenances.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Sedimentary rocks forming processes; Classification and characterization of rudaceous, arenaceous, argillaceous and calcareous rocks; Primary and secondary sedimentary structures; Mineralogical attributes, textural properties and diagenetic processes of sedimentary rocks.				15
II	Petrography of major siliciclastic and carbonate rocks namely conglomerate, breccia, quartzarenite, arkose, lithic arenite, quartzwacke, feldspathicwacke, lithicwacke, mudrocks / shale, limestones: crystalline, micritic and sparitic.				15
III	Heavy minerals analyses and their importance in provenance interpretations; Processes and tectonic settings of various depositional environments like fluvial, Deltaic, coastal and deep-marine.				15
Texts / References	1. Ehlers, E. G., & Blatt, H. (1982). <i>Igneous, Sedimentary, and Metamorphic Petrology</i>. CBS Publ. 2. Greensmith, J. T. (1984). <i>Petrology of Sedimentary Rocks</i>. Thomas Murby Publ. 3. Hatch, F. H., Rastall, R. H., & Black, M. (1938). <i>Petrology of Sedimentary Rocks</i>. Thomas Murby Publ. 4. Pettijohn, F. J. (1957). <i>Sedimentary Rocks</i> (3rd ed.). Oxford Book Co. 5. Donald, R. P., & Schwab, F. (1996). <i>Sedimentary Geology</i>. W. H. Freeman and Company, New York. 6. Boggs, S. (2006). <i>Principles of Sedimentology and Stratigraphy</i>. Prentice Hall.				
Learning Outcomes (not more than 100 words)	Through this course, the students will learn about the various sedimentary processes and products. Moreover, they will learn about the rock types, their textures and mineralogy, details of sedimentary structures and diagenesis. They will also learn about the sources of various sediments and their depositional environments.				

Course Title	ESMJ53:Igneous Petrology				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	03	---	03	
	Hour of Teaching (Total)	45	---	45	
Course Objectives (not more than 100 words)	The objective of the Igneous Petrology course is to provide students with an in-depth understanding of the origin, classification, and evolution of igneous rocks. The course aims to equip students with the skills to interpret igneous textures, understand phase equilibria, and apply classification systems. Additionally, it seeks to develop students' ability to analyze a variety of igneous rocks within the context of their petrogenesis, with an emphasis on Indian examples.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Magma: Its physics, nature, factors affecting magma and its evolution. Thermal structure of the earth and melting of mantle. Plate tectonics and generation of different magmas in various tectonic settings.				08
II	Concept of primary and secondary magma. Magma series, dynamics, differentiation, emplacement, and crystallization of the magma. Magma mixing, mingling, and immiscibility. Plume magmatism and hotspots. Mantle metasomatism. Mantle heterogeneities.				07
III	Partial melting (batch and fractional melting), crystal fractionation (equilibrium and fractional (Rayleigh) crystallization), contamination (AFC process) and dynamic melting.The phase equilibrium of binary (Ab-An, Ab-Or, Di-An, Fo-Si) and ternary (Di-Ab-An, Di-Fo-Si, Di-Fo-An, Ne-Ks-Si, Fo-An-Si) systems and their relation to magma genesis and crystallization in the light of modern experimental works.				15
IV	IUGS classification of the igneous rocks. CIPW Norm. Petrology and petrogenesis of the following igneous rocks with suitable Indian examples: (i) Peridotites, komatiites, gabbros, basalts, anorthosites, ophiolites. (ii) Large Igneous Provinces, mafic dyke swarms, boninites, and layered complexes. (iii) Alkaline rocks, carbonatites, kimberlites, lamproites, kamafugites, and lamprophyres. (iv) Granitoids, adakites, and sanukitoids.				15
Texts / References	1. Winter, J. D. (2010).<i>Principles of Igneous and Metamorphic Petrology</i>. Pearson Prentice Hall. 2. Gill, R. (2010).<i>Igneous Rocks and Processes: A Practical Guide</i>. John Wiley & Sons. 3. Sen, G. (2014).<i>Petrology: Principles and Practice</i>. Springer-Verlag. 4. Philpotts, A., & Ague, J. (2009).<i>Principles of Igneous and Metamorphic Petrology</i>. Cambridge University Press. 5. Cox, K. G., Bell, J. D., & Pankhurst, R. J. (1979).<i>The Interpretation of Igneous Rocks</i>. Chapman and Hall Publishing. 6. Wilson, M. (1989).<i>Igneous Petrogenesis: A Global Tectonic Approach</i>. Chapman and Hall Publishing. 7. Frost, B. R., & Frost, C. D. (2014).<i>Essentials of Igneous and Metamorphic Petrology</i>. Cambridge University Press.				
Learning Outcomes (not more than 100 words)	By completing this course, students will gain a comprehensive understanding of igneous rocks and the processes leading to their formation. They will learn about the physics and nature of magma, including its differentiation and crystallization, the Earth's thermal structure, mantle melting, and the role of plate tectonics in magma generation. Students will understand primary and secondary magmas, magma dynamics, and the significance of plume magmatism, hot spots, and mantle heterogeneities. They will explore phase equilibria in binary and ternary systems, interpret igneous textures, and apply IUGS classification to identify igneous rocks, focusing on Indian examples.				

Course Title	ESMJ54:Mineralogy and Crystallography				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	03	---	03	
	Hour of Teaching (Total)	45	---	45	
Course Objectives (not more than 100 words)	1. To get acquainted with the various silicate structures of rock forming minerals and mineral groups. 2. To identify the structural formulae, classification, atomic structure, polymorphic/structural transformations and related processes in silicate minerals. 3. To learn the various principles of Optical crystallography with the application of the concepts of optical mineralogy for uniaxial and biaxial minerals. 4. To learn about the 32 crystals classes belonging to 7 crystal systems along with the description of the different normal classes and about the development of X ray crystallography. 5. To get acquainted with the applications of Electron Micro Probe analyses and Scanning electron Microscopy in mineral sciences.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	An in-depth examination of the structural categories of silicate minerals. Detailed methodologies for preparing thin sections used in mineralogical analysis. Comprehensive Study of Significant Silicate Mineral Groups: Nesosilicates (Olivine, Garnet, Aluminosilicate); Cyclosilicates (Beryl); Inosilicates (Pyroxene, Amphibole); Phyllosilicates (Kaolinite, Serpentine, Mica); Tectosilicates (Feldspar, Feldspathoid). Aspects to be covered for each mineral groups: General and structural formulae; Classification and atomic structure; Polymorphism and structural transformations; Isomorphism and pseudomorphism; Chemical composition including elemental substitutions and solid solution mechanisms; Experimental data on pressure-temperature stability conditions; Modes of occurrence and alteration processes.				15
II	Study of light interaction with crystals exhibiting one and two optic axes. Indicatrix Construction and Analysis: Understanding the geometric representation of refractive indices in different crystal systems. Examination of colour variation in minerals when viewed under polarized light from different orientations. Analysis and interpretation of interference patterns to determine optical properties and orientations of crystals. Measurement and significance of the optic axial angle (2V) and extinction angle (2E) in mineral identification.				10
III	Classification of Crystal Systems and characterization of all 32 crystal classes and their respective normal classes. Crystal Projection Techniques (Spherical Projection & Stereographic Projection) with applications and utility of projection methods in crystallographic studies. Twinning and Twin Laws: Exploration of common twin types, associated twin laws, and examples of twinning phenomena in various minerals. Overview of the 14 Bravais lattices and their significance in defining crystal structures.				10
IV	Tracing the evolution and milestones in the field of X-ray crystallography. Derivation and theoretical underpinnings of Bragg's Law. Application of Bragg's Law in determining crystal structures through X-ray diffraction methods. Utilization of X-ray in mineral science. Principles and methodologies of electron microprobe analysis (EMPA) in quantitative mineral chemistry. Application of scanning electron microscopy (SEM) for detailed morphological and compositional studies of mineral specimens.				10
Texts References	1. Berry, L. G., Mason, B., & Dietrich, R. V. (1985). <i>Mineralogy: Concepts, Descriptions, and Determinations</i> . CBS Publishers. 2. Dana, E. S., & Ford, W. E. (2002). <i>A Textbook of Mineralogy</i> (Reprint). 3. Deer, W. A., Howie, R. A., & Zussman, J. (2013). <i>An Introduction to the Rock-Forming Minerals</i> . ELBS and Longman. 4. Gribble, C. D. (2005). <i>Rutley's Elements of Mineralogy</i> . Springer. 5. Kerr, P. F. (1977). <i>Optical Mineralogy</i> . McGraw Hill. 6. Nesse, D. W. (1986). <i>Optical Mineralogy</i> . McGraw Hill.				

	7. Perkins, D. (2013). <i>Mineralogy</i> . Prentice Hall. 8. Phillips, F. C. (1971). <i>Introduction to Crystallography</i> . Longman Group Publication. 9. Verma, P. K. (2010). <i>Optical Mineralogy</i> . CRC Press. 10. Sharma, R. S., & Sharma, A. (2013). <i>Crystallography and Mineralogy</i> . Geological Society of India. 11. Reed, S. J. B. (1996). <i>Electron Microprobe Analysis and Scanning Electron Microscopy in Geology</i> . Cambridge University Press.
Learning Outcomes (not more than 100 words)	1. The learners would be able to identify and differentiate between various silicate structures found in rock forming minerals. 2. The learners would be able to appraise the role of various elemental and structural transformations in silicate minerals. 3. The learners would be able to understand various crystallographic properties of uniaxial and biaxial crystals based on optical parameters like Indicatrix, pleochroism, Interference figures, crystal orientation, 2V and 2E angles. 4. The learners would be able to apply the concept of symmetry elements w.r.t. 32 crystal classes and twinning in crystals. 5. The learners would be able to understand the importance and application of EPMA and SEM in understanding various physico- chemical characteristics of minerals.

Course Title	ESMJ55: Practical connected with ESMJ -51 and ESMJ -52			
Category of Course	Major			
Credits&Hour of Teaching		Theory	Practical	Cumulative
	Credits	---	01	01
	Hour of Teaching (Total)	---	30	30
Course Objectives (not more than 100 words)	To impart the knowledge about the various metamorphic and sedimentary rocks in terms of their megascopic and microscopic details. To provide knowledge on classifying metamorphic rocks into facies, subfacies, and zones using graphical representation techniques (ACF, AKF, and AFM diagrams). To infer the genesis of sedimentary rock types from various sedimentary processes such as clast generation, chemical and biochemical.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Study of metamorphic rocks of different metamorphic facies in Hand Specimen - slate, phyllite, schist, gneiss, marble, quartzite, charnockite, hornfels, khondalite			10
II	Study of Metamorphic Rocks (slate, phyllite, schist, gneiss, marble, quartzite, charnockite, hornfels, khondalite) in thin sections belonging to different facies with emphasis on texture/structure, mineral composition, parent rock, metamorphic facies / subfacies / zone to which the rock can be assigned and graphical representation of the assemblage in ACF, AKF and AFM diagrams.			10
III	Megascopic examination of conglomerate, breccia, quartz arenite, arkose, lithic arenite, quartzwacke, feldspathicwacke, lithicwacke(grewacke), mudrocks/shale and carbonates (micrite, sparite and accretionarylimestones).			07
IV	Examination of some common heavy minerals in grain mounts			03
Texts References	/ 1. Kerr, P.F., (1959). <i>Optical Mineralogy</i> . McGraw-Hill Book Company Inc., New York, pp.442. 2. Mackenzie, W.S., and Guilford, C., (1993). <i>Atlas of Rock-Forming Minerals in Thin Section</i> . Longman Scientific and Technical, EnESand, pp. 98. 3. Moorhouse, W.W., (1985). <i>The Study of Rocks in Thin Section (Indian Edition)</i> . CBS Publications and Distributors, Delhi, pp. 514. 4. Passchier, C.W., and Trouw, R.A.J., (1996) <i>Microtectonics</i> . Springer – Verlag Publications, pp. 289 (with 254 figures). 5. Phillips, R. W., and Griffen, D.T., (1986). <i>Optical Mineralogy – The Non-opaque Minerals (Indian Edition)</i> . CBS Publishers and Distributors, Delhi, pp. 677. 6. Spry, A. (1976). <i>Metamorphic Textures</i> , Pergamon Press.			

	7. Williams, H., Turner, F.J. and Gilbert, C.M., (Indian Edition, 1985). Petrography: An Introduction to the Study of Rocks in Thin Sections. CBS Publishers and Printers, Delhi, pp. 626. 8. Yardley, B.W.D., MacKenzie, W.S., and Guilford, C., (1995). Atlas of Metamorphic Rocks and Their Textures. Longman Scientific and Technical, EnESand, pp. 120. 9. Collins, J.D., and Thompson, D.B., (1982). Sedimentary structures. George Allen and Unwin, London. 10. Lindholm, R.C., (1987). A practical approach to sedimentology. Allen and Unwin, London. 11. Tucker, M.E., (1981). Sedimentary Petrology: An introduction. Wiley and Sons, New York.
Learning Outcomes (not more than 100 words)	The students will be skilled in identifying various metamorphic and sedimentary rocks in the field as well as in the laboratory. This will help them in visualizing their origin as sub-surface and surface processes.

Course Title	ESMJ 56: Practical connected with ESMJ -53 and ESMJ -54			
Category of Course	Major			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	---	01	01
	Hour of Teaching (Total)	---	30	30
Course Objectives (not more than 100 words)	The objective of the course is to equip students with hands-on skills in analyzing igneous rocks and minerals. Students will learn to identify rock-forming minerals and Igneous rocks in hand specimens and thin sections. Additionally, students will understand and represent the symmetry elements of crystals in 32 symmetry classes and their stereograms, preparing them for advanced geological research. In general, through practical exercises, students will gain a deeper understanding of geological processes and improve their analytical and observational skills, preparing them for advanced studies in geosciences.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Megascopic and microscopic study of different igneous rocks. Calculation of CIPW Norms. Identification of rock-forming minerals in hand specimens. Determination of length-fast and length-slow characters of minerals. Scheme of pleochroism and absorption of a given mineral in thin section. Determination of extinction angles and composition of plagioclase. Study of interference figures of uniaxial and biaxial crystals, determination of optic signs. Goniometer and its use in measuring interfacial angles of crystals and calculation of axial ratio. Representation of symmetry elements of crystals belonging to 32 classes of symmetry and study of their stereograms.			30
Texts / References	1. J. D. Winter (2010).<i>Principles of Igneous and Metamorphic Petrology</i>. Pearson Prentice Hall. 2. Robin Gill (2010).<i>Igneous Rocks and Processes: A Practical Guide</i>. John Wiley & Sons. 3. Gautam Sen (2014).<i>Petrology: Principles and Practice</i>. Springer-Verlag Publisher. 4. A. Philpotts and J. Ague (2009).<i>Principles of Igneous and Metamorphic Petrology</i>. Cambridge University Press. 5. K. G. Cox, J. D. Bell, and R. J. Pankhurst (1979).<i>The Interpretation of Igneous Rocks</i>. Chapman and Hall Publishing. 6. M. Wilson (1989).<i>Igneous Petrogenesis: A Global Tectonic Approach</i>. Chapman and Hall Publishing. 7. B. R. Frost and C. D. Frost (2014).<i>Essentials of Igneous and Metamorphic Petrology</i>. Cambridge University Press. 8. Berry, L. G., Mason, B., and Dietrich, R. V. (1985).<i>Mineralogy: Concepts, Descriptions, and Determinations</i>. CBS Publishers. 9. Dana, E. S., and Ford, W. E. (2002).<i>A Textbook of Mineralogy</i> (Reprint). 10. Deer, W. A., Howie, R. A., & Zussman, J. (2013).<i>An Introduction to the Rock-Forming</i>			

	<i>Minerals</i>. ELBS and Longman. 11. Gribble, C. D. (2005).<i>Rutley's Elements of Mineralogy</i>. Springer. 12. Kerr, P. F. (1977).<i>Optical Mineralogy</i>. McGraw Hill. 13. Nesse, D. W. (1986).<i>Optical Mineralogy</i>. McGraw Hill. 14. Perkins, D. (2013).<i>Mineralogy</i>. Prentice Hall. 15. Phillips, F. C. (1971).<i>Introduction to Crystallography</i>. Longman Group Publication. 16. Verma, P. K. (2010).<i>Optical Mineralogy</i>. CRC Press. 17. Sharma, R. S., and Sharma, A. (2013).<i>Crystallography and Mineralogy</i>. Geological Society of India. 18. Reed, S. J. B. (1996).<i>Electron Microprobe Analysis and Scanning Electron Microscopy in Geology</i>. Cambridge University Press.
Learning Outcomes (not more than 100 words)	By completing this course, students will develop expertise in identifying and analyzing igneous rocks and minerals both megascopically and microscopically. They will master the calculation of CIPW norms and the determination of mineral optical properties such as pleochroism, extinction angles, and optic signs. Students will gain proficiency in using a goniometer to measure interfacial angles and calculate axial ratios. Additionally, they will understand the symmetry elements of crystals and their representation in stereograms. This course will enhance their practical skills in mineralogy and crystallography, preparing them for advanced geological research.

Course Title	ESMV51: Geoarcheology and Museology			
Category of Course	Minor			
Credits& Hour of Teaching		Theory	Practical	Cumulative
	Credits	02	02	04
	Hour of Teaching (Total)	30	60	90
Course Objectives (not more than 100 words)	This course aims to introduce students to the fundamental principles and practices of geoarchaeology and museology. Students will learn geological methods and applications in archaeology, including site analysis, dating techniques, and case studies. The course also covers curatorial practices, artifact conservation, museum management, and emerging technologies in museology. Emphasis is placed on understanding the interplay between geology and archaeology, and the role of museums in preserving and presenting historical artifacts.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Geoarchaeology: Introduction to geoarchaeology; Geological applications in archaeology; Methods in geoarchaeology; Post-depositional processes of archaeological sites; Pedomstratigraphy and archeostratigraphy in geoarchaeology;			10
II	Paleoclimate, paleovegetation, human evolution and migration; Geomorphology; Dating methods; Geoarchaeological case studies.			10
III	Museology: Introduction to museology; Curatorial practices and artifact conservation; Outsourcing of archives; Management of museums; Emerging technologies in museology; Future trends and collaboration.			10
IV	Fieldwork in local geoarchaeological sitesLaboratory techniques in geoarchaeology: sediment processing for grain size analysis, microfossils, and mineralogy.Visit to the local museum or cultural heritage site; Preparation of report of the museum visit; Cataloguing (inventory and documentation) of specimens in the museum.			60
Texts / References	Geoarchaeology: 1. Rapp, G.R., Hill, C.L., and Hill, M.C.L., 2006. <i>Geoarchaeology: The Earth-Science Approach to Archaeological Interpretation</i>. Yale University Press. 2. Cordova, C., 2018. <i>Geoarchaeology: The Human-Environmental Approach</i>. Bloomsbury Publishing. 3. Pitul'ko, V.V., and Pavlova, E.Y., 2016. <i>Geoarchaeology and Radiocarbon Chronology of Stone Age Northeast Asia</i>. Texas A&M University Press. 4. Dincauze, D.F., 2000. <i>Environmental Archaeology: Principles and Practice</i>. Cambridge University Press.			

	5. Waters, M.R., 1992. Principles of Geoarchaeology: A North American Perspective. University of Arizona Press. 6. Goldberg, P., Macphail, R., Carey, C., & Zhuang, Y., (2022). Practical and Theoretical Geoarchaeology (2nd ed.). Wiley. Museology: 1. Ambrose, T., and Paine, C., 2006. Museum Basics: The International Handbook. Routledge. 2. Matassa, F., 2011. Museum Collections Management. Facet Publishing. 3. Macdonald, S., ed., 2011. A Companion to Museum Studies. John Wiley & Sons. 4. Imam, J.Md., 2018. Introduction to Museology. World Heritage Inc. 5. Pearce, S.M., ed., 1994. Interpreting Objects and Collections. Psychology Press. 6. Weil, S., 2012. Making Museums Matter. Smithsonian Institution.
Learning Outcomes (not more than 100 words)	Upon completion of this course, students will be able to apply geological methods to archaeological research, including site analysis and dating techniques They will analyze the role of paleoclimate and geomorphology in human evolution and migration, and implement curatorial practices and conservation techniques for artifact preservation. Additionally, students will manage museum operations, utilize emerging technologies in museology, and evaluate case studies to identify successful geoarchaeological and museological practices. They will also be equipped to collaborate effectively in multidisciplinary teams to address challenges in archaeology and museum studies.

Course Title	ESIN-51: (Internship): [Students can do internship after first to fourth semester during summer vacation]			
Category of Course	Internship			
Credits& Hour of Teaching		Theory	Practical	Cumulative
	Credits	---	02	02
	Hour of Teaching (Total)	---	60	60
Course Objectives (not more than 100 words)	The course objectives of an internship in geology typically focus on providing students with practical, hands-on experience that complements their academic studies. Here are some common objectives for a geology internship course: 1. Develop practical skills in fieldwork, including data collection, geological mapping, and rock and mineral sampling. 2. Gain experience with laboratory analyses and techniques used to study geological samples and data.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Internshipistobedoneduringvacation			60
Texts / References				
Learning Outcomes (not more than 100 words)	The learning outcomes of an internship in geology are to reflect the knowledge, skills, and experiences gained through hands-on, practical engagement in the field. 1. It will demonstrate proficiency in fieldwork techniques, including geological mapping and sample collection. 2. It will provide hands-on experience with laboratory analyses and interpret the results accurately.			

Semester: VI

Course Title	ESMJ61: Structural Geology				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	03	---	03	
	Hour of Teaching (Total)	45	---	45	
Course Objectives (not more than 100 words)	The course focuses on advanced level knowledge on selected topics of structural geology and associated laws of physics related to deformation behavior of rocks.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Examination of the properties of rocks and the factors that influence them. Introduction to the concept of stress and two-dimensional stress analysis. Overview of rock failure theories, with a focus on the causes and dynamics of fracturing and faulting.				10
II	Investigation into the mechanics and causes of folding. Genetic classification of folds and analysis of fold superposition patterns. Study of strain distribution within folds and its implications.				5
III	Detailed analysis of two-dimensional strain, including the properties and geological significance of strain ellipses and ellipsoids. Identification of strain markers and methods for measuring strain in naturally deformed rocks.				10
IV	Introduction to fold-thrust belts, including the concepts of thin- and thick-skinned deformation, decollement. Introduction to shear zones (both brittle and ductile). Analysis of the geometry and products of shear zones, with a focus on mylonites and cataclastites.				10
V	Exploration of the origin and significance of planar and linear fabrics in deformed rocks. Application of stereographic and equal-area projections in structural analysis, including the use of π and β diagrams. Geometric analysis of simple and complex macroscopic structures, with an introduction to petrofabric analysis.				10
Texts / References	1. Davis, G.H., Reynolds, S.J., and Kluth, C.F. (2012). Structural Geology of Rocks and Regions (3rd Ed.). Wiley. 2. Fossen, H. (2016). Structural Geology (2nd Ed.). Cambridge University Press. 3. Hobbs, B.E., Means, W.D., and Williams, P.F. (1976). An Outline of Structural Geology. John Wiley and Sons, New York. 4. Ramsay, J.G. (2004). Folding and Fracturing of Rocks. The Blackburn Press. 5. Ramsay, J.G., and Huber, M.I. (1983). Techniques of Modern Structural Geology Vol. I and Vol. II. Academic Press. 6. Twiss, R.J., and Moores, E.M. (2006). Structural Geology (Second Edition). W.H. Freeman.				
Learning Outcomes (not more than 100 words)	The student is expected to know about the concept of geological stress, strain and various theories to explain the behavior of rocks in ideal and non-ideal conditions. The student will also learn where the various types of deformational structure would be found and how to study them under natural conditions				

Course Title	ESMJ62: Economic Geology				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	03	---	03	
	Hour of Teaching (Total)	45	---	45	
Course Objectives (not more than 100 words)	To introduce the basic concept of mineral economics, ore forming processes and classification of economic mineral deposits. To introduce fundamental aspects of origin, mode of occurrences and the distribution of economic mineral deposits in India.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Fundamental of economic geology: minerals, ores, and gangues; Tenor of ores; Common mode of occurrences, forms and structures of mineral deposits; Basic concepts of ore-genesis, paragenesis, paragenetic sequence and zoning in metallic ore deposits, Importance of economic geology.				10
II	Ore forming minerals, Processes of formation of ore deposits; origin and mode of occurrences of metallic mineral deposits of iron, manganese, gold, copper, aluminium, lead, zinc, chromite, magnesium, tin, tungsten, with their economic uses and Indian Distribution.				10
III	Processes of formation of mineral deposits; Concepts on Magmatic processes, Hydrothermal processes, Metamorphic processes and Sedimentary or surficial processes.				10
IV	Origin and Mode of occurrences of Non-metallic mineral deposits with particular reference to the industries (Cement, Glass, Ceramics, Refractory, Fertilizer, Chemicals, Abrasive and Gemstones) and their Indian distribution.				10
V	Origin and processes of formation, mode of occurrence, economic uses and distribution of coal and petroleum deposits in India; Detail geological study of atomic fuels deposits of uranium, thorium and rare earth elements deposits in India.				05
Texts / References	1. Mead L. Jense, and Alan M. Bateman (2013) Economic Mineral Deposits. Third Edition, John Wiley and Sons. 2. AshokeMookherjee (2000): Ore Genesis-A Holistic Approach, Allied Publisher. 3. Umeshwar Prasad (2019) Economic Geology. CBS Publishers & Distributors, New Delhi. 4. Anthony M. Evans (2006) Ore Geology and Industrial Minerals: An Introduction. Third Edition, Blackwell Science. 5. R. K. Sinha and N. L. Sharma (2019) Mineral Economics. Fourth Edition, Oxford & IBH Publishing. 6. Sanjay Kumar Tiwari (2019) Ore Geology, Economic Minerals and Mineral Economics. Atlantic Publishers & Distributors Pvt Ltd. 7. Walter L Pohl (2010) Economic Geology: Principles and Practice. First Edition, John Wiley. 8. Mihir Deb and Sanjib Chandra Sarkar (2017) Minerals and Allied Natural Resources and their Sustainable Development. Springer Singapore. 9. John Ridley (2013) Ore Deposit Geology. Cambridge University Press. 10. Laurence Robb (2005) Introduction to Ore-Forming Processes. John Wiley & Sons.				
Learning Outcomes (not more than 100 words)	Upon successful completion, students will have the knowledge and skills to: Recognize common ore and economic minerals in hand specimens Demonstrate knowledge with a wide range of mineral deposits, including recognizing the overall geometry, zonation and alteration patterns associated with specific classes of metallic mineral deposits. Relate overall geometry, zonation and alteration patterns of rock associations to specific classes of metallic mineral deposits. Evaluate different geological processes of element enrichment to from economic mineral deposits.				

Course Title	ESMJ63: Paleontology				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	03	---	03	
	Hour of Teaching (Total)	45	---	45	
Course Objectives (not more than 100 words)	To understand the history of evolution of life for better comprehension of geodynamic evolution of the earth through geological time.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Applicative dimensions of paleontology in geology; Index fossils; Modern concept of the origin of life; Lamark's and Darwin's theories of organic evolution with suitable evidences.				10
II	Sampling and processing techniques in macro and micro-fossil; System of binomial nomenclature in taxonomic studies; Species concept; Composition of hard parts; Growth of individual organism.				10
III	A detailed account of diagnostic morphological features and geological distribution of the invertebrate fossil groups as under: Brachiopoda, Bivalvia, Gastropoda, Cephalopoda; Stratigraphic implications of ammonites and conodonts; Microfossils and their different kinds.				15
IV	The study of various ways of preservation and broad features of plant fossils; Preliminary ideas of Gondwana flora; Elementary knowledge about vertebrate classes and significance of Indian vertebrate fauna.				10
Texts References	1. Black, R.M., (1988). The Elements of Palaeontology, Cambridge Univ. 2. Clarkson, E.N.K., (1986). Invertebrate Palaeontology and Evolution, Allen and Unwin Publ. 3. Jain, P.C. and Anantharaman, M.S., (1983). Palaeontology: Evolution and Animal Distribution, Vishal Publ. 4. Lehmann, U., (1983). Fossils Invertebrate, Cambridge Univ. Press. 5. Benton, Michael J. (2014). Vertebrate Paleontology. Blackwell Publ., Chapman and Hall. 6. Benton, Michael J. and Harper, David A.T., (2020). Introduction to Paleobiology and the Fossil Record. John Wiley & Sons Ltd. 7. Foote, M. and Miller, A.I., (2007). Principals of Paleontology. John Wiley & Sons Ltd. 8. Moore, R.C., Lalicker, C.G. and Fischer, A.G., (1997). Invertebrate Fossils, CBS Publ. 9. Nield, E.W. and Tucker, V.C.T., (1985). Palaeontology: An Introduction, Pergmon Press. 10. Prothero, D.R., (2004). Bringing Fossil to Life – An Introduction to Paleontology (2nd Ed.), McGraw Hill. 11. Shrock, R.R. and Twenhoffel, W.H., (1952). Principles of Invertebrate Paleontology, CBS Publ. 12. Raup, D.M. and Stanley, S.M., (1985). Principles of Palaeontology, CBS Publ. 13. Stebbins, (1979). Process of Organic Evolution (3rd Ed.) Prentice Hall. 14. Woods, H., (1985). Palaeontology Invertebrate, CBS Publ. 15. Doyle, P., (2014). Understanding fossils, An introduction to Invertebrate Paleontology. John Wiley & Sons.				
Learning Outcomes (not more than 100 words)	The study of palaeontology plays crucial role in the development of high resolution biostratigraphy, assigning the ages to the rock successions and their global correlations. Such study is of great significance in understanding palaeogeography, palaeoenvironment, palaeodeposition and palaeoclimate. This study will enhance the geological concepts of students essential for any kind of geological exploration.				

Course Title		ESMJ64: Stratigraphy			
Category of Course		Major			
Credits& Hour of Teaching			Theory	Practical	Cumulative
		Credits	03	---	03
		Hour of Teaching (Total)	45	---	45
Course Objectives (not more than 100 words)		The course focuses on concept of stratigraphy and geologically important stratigraphic boundaries in India and their equivalents.			
Course Content		The course content is divided into minimum 4 units along with defined hours of teaching for each unit.			
	Course Content				Hr. of Teaching
I	Concept of facies, refinementand correlation in stratigraphy;Conceptof sequence stratigraphy and paleobiogeography; A brief introduction of Bio-magneto-seismic-chemo-andeventstratigraphy.				11
II	CuddapahandVindhyan supergroupsandtheirequivalents,Precambrian/Cambrianboundary. Stratigraphic sequence of the Paleozoic of Kashmir and Gondwana Supergroup,				11
III	Permian/TriassicBoundaryinIndia.A detailed outline of Lithostratigraphic framework and fossil content of Triassic of Spiti, Jurassic of Kachchh and Cretaceous of Tiruchirapalli.				11
IV	Cretaceous/PaleogeneBoundaryinIndia.The lithostratigraphy and fossil content of the following: Paleogene of Assam,Kachchh and Andaman, Siwaliks of Jammu and Himachal Pradesh and KarewasofKashmir.Stratigraphic study of Deccan, Rajmahal Trap and Intertrappean beds.				12
Texts / References		1. Boggs, S. (2001). Principles of Sedimentology and Stratigraphy. Prentice Hall. 2. Dunbar, C.O. and Rodgers, J. (1957). Principles of Stratigraphy. John Wiley and Sons. 3. Krishnan, M.S. (1968). Geology of India and Burma. Higgibothon, Madras. 4. Kumar, R. (1985). Historical Geology and Stratigraphy of India. Wiley Eastern Ltd. 5. Wadia, D.N. (1966). Geology of India. English Language Publ. 6. Kumar, R. (2022). Fundamentals of Historical Geology and Stratigraphy of India. New Age International Publishers. 7. Brookfield, M.E. Principles of Stratigraphy. Wiley Blackwell. 8. Weller, J.M. (1960). Stratigraphic Principles and Practices. Universal Book. 9. Valdiya, K.S. (2016). Geodynamic Evolution of India (2nd Edition). Springer. 10. Krishna, J. (2017). The Indian Mesozoic Chronicle: Sequence Stratigraphic Approach. Springer, Singapore. 11. Ramakrishna, M. and Vaidyanadhan, R. (2010). Geology of India (Vol. 1 & 2). Geological Society of India, Bangalore. 12. Sharma, R.S. (2010). Cratons and Fold Belts of India. Springer. 13. Chetty, T.R.K. (2018). Proterozoic Orogens of India: A Critical Window to Gondwana. Elsevier. 14. Sarkar, S.C. and Gupta, A. (2012). Crustal Evolution and Metallogeny in India. Cambridge University Press. 15. Doyle, P. Understanding Fossils: An Introduction to Invertebrate Paleontology. John Wiley & Sons. 16. Prothero, D.R. Bringing Fossils to Life: An Introduction to Paleobiology. Columbia University Press, New York.			
Learning Outcomes (not more than 100 words)		After going through these exercises, the students will be able to know different types of boundary problems in India and Their equivalents. They will also able to learn about the geologically important stratigraphic rocks associated with different Supergroups in India.			

Course Title	ESMJ65: Practical connected with ESMJ61 and ESMJ62				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	---	01	01	
	Hour of Teaching (Total)	---	30	30	
Course Objectives (not more than 100 words)	The course focuses on advanced level practical exercises on structural geology and economic geology.Practical knowledge to recognize the most common ore and economic minerals used in the industry, their compositions and be able to identify them based on their macroscopic and physical characteristics in hand specimen.Knowledge of Indian distribution of economic mineral deposits and their industrial applications.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Practice problems focused on solving complex orthographic challenges, with an emphasis on interpreting and extracting detailed information from intricate geological maps.				8
II	Advanced-level exercises utilizing stereographic nets for the analysis and interpretation of geological data, including orientation and structural relationships.Practice in calculating strain from geological strain markers, focusing on advanced techniques and methodologies for quantifying deformation in natural rock formations.				7
III	Economic Geology:Study of ore and economic minerals in hand specimen of iron ores, manganese ores, bauxite, galena, sphalerite, chromite, magnesite, wolframite and cassiterite, etc.				8
IV	Preparation of maps showing distribution of important metallic and non-metallic, industrial minerals deposits;Distribution of important atomic fuels, coal and oil fields of India.				7
Texts / References	Structural Geology: - Groshong, R.H. and Groshong, R.H. (2006). 3-D Structural Geology. Springer-Verlag Berlin Heidelberg. - Lisle, R.J. and Leyshon, P.R. (2004). Stereographic Projection Techniques for Geologists and Civil Engineers (2nd edition). Cambridge University Press. - Lisle, R.J. (2020). Geological Structures and Maps: A Practical Guide. Butterworth-Heinemann. - Marshak, S. and Mitra, G. (1988). Basic Methods of Structural Geology. Prentice Hall. - Ragan, D.M. (2009). Structural Geology: An Introduction to Geometrical Techniques (Fourth Edition). Cambridge University Press. - Rowland, S.M., Duebendorfer, E.M. and Schiefelbein, I.M. (2007). Structural Analysis and Synthesis: A Laboratory Course in Structural Geology (3rd edition). Wiley-Blackwell. Economic Geology: - Mookherjee, A. (2000). Ore Genesis: A Holistic Approach. Allied Publisher. - Prasad, U. (2019). Economic Geology. CBS Publishers & Distributors, New Delhi. - Deb, M. and Sarkar, S.C. (2017). Minerals and Allied Natural Resources and Their Sustainable Development. Springer Singapore. - Singh, Y. (2020). Rare Earth Element Resources: Indian Context. Springer Cham. - Aswathanarayana, U. (1985). Principles of Nuclear Geology. Routledge, 1st edition.				
Learning Outcomes (not more than 100 words)	The student will be learning the advanced level and complex structural geological problems. Practical connected with hand specimen studies of the economic minerals as a practice of identifying minerals with their characters. Maps showing distribution of occurrences of such minerals will also be practiced. Upon successful completion, students will have the knowledge and skills to: Recognize common ore and economic minerals in hand specimens Demonstrate knowledge with a wide range of mineral deposits in India specific classes of metallic, industrial, ceramic, cement and fuel mineral deposits.				

Course Title	ESMJ66: Practical connected with ESMJ63 and ESMJ64				
Category of Course	Major				
Credits & Hour of Teaching		Theory	Practical	Cumulative	
	Credits	---	01	01	
	Hour of Teaching (Total)	---	30	30	
Course Objectives (not more than 100 words)	There will be practical exercises to learn about detail morphological features of different types of fossils. Students will also be able to learn about the different sedimentary basins including land/sea distribution in reference to India.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Paleontology: Study of morphological characters, systematic positions, and age of about 30 representative genera belonging to the following groups: Brachiopoda, Bivalvia, Cephalopoda, and Gastropoda. A detailed systematic description of the following genera: Rhynchonella, Terebratula, Arca, Modiolus, Perisphinctes, Nautilus, Natica, and Conus.				10
II	Stratigraphy: Distribution of the following geological formations on a sedimentary basin map of India: Cuddapah Supergroup, Vindhyan Supergroup, Marine Lower Permian, Gondwana Supergroup, Marine Mesozoics, Deccan Traps and equivalents, Marine Cenozoic, and Siwalik Group.				10
III	Preparation of land/sea distribution on sedimentary basin map of India during Late Precambrian/Early Cambrian, Early Permian, Jurassic, Cretaceous, and Eocene; Study of rocks from important Indian stratigraphic horizons. Some selected exercises on stratigraphic classification.				10
Texts / References	1. Black, R. M. (1988): <i>The Elements of Palaeontology</i>, Cambridge University Press. 2. Clarkson, E. N. K. (1986): <i>Invertebrate Palaeontology and Evolution</i>, Allen and Unwin Publishers. 3. Jain, P. C. and Anantharaman, M. S. (1983): <i>Palaeontology: Evolution and Animal Distribution</i>, Vishal Publishers. 4. Lehmann, U. (1983): <i>Fossils Invertebrate</i>, Cambridge University Press. 5. Michael J. Benton & David A. T. Harper (2020): <i>Introduction to Paleobiology and the Fossil Record</i>, John Wiley & Sons Ltd. 6. Michael Foote & Arnold I. Miller: <i>Principles of Paleontology</i>, John Wiley & Sons Ltd. 7. Moore, R. C., Lalicker, C. G. & Fischer, A. G. (1997): <i>Invertebrate Fossils</i>, CBS Publishers. 8. Nield, E. W. & Tucker, V. C. T. (1985): <i>Palaeontology: An Introduction</i>, Pergamon Press. 9. Prothero, D. R. (2004): <i>Bringing Fossils to Life – An Introduction to Paleontology</i> (2nd Ed.), McGraw Hill. 10. Rastogi (1988): <i>Organic Evolution</i>, Kedarnath and Ramnath Publishers. 11. Raup, D. M. & Stanley, S. M. (1985): <i>Principles of Palaeontology</i>, CBS Publishers. 12. Shrock, R. R. & Twenhoffel, W. H. (1952): <i>Principles of Invertebrate Paleontology</i>, CBS Publishers. 13. Stebbins (1979): <i>Process of Organic Evolution</i> (3rd Ed.), Prentice Hall. 14. Woods, H. (1985): <i>Palaeontology Invertebrate</i>, CBS Publishers. 15. Peter Doyle: <i>Understanding Fossils, An Introduction to Invertebrate Paleontology</i>, John Wiley & Sons. 16. Donald R. Prothero: <i>Bringing Fossils to Life; An Introduction to Paleobiology</i>, Columbia University Press, New York.				
Learning Outcomes (not more than 100 words)	The student will be learning the morphological features of geologically important fossils. Practical connected with hand specimen studies of the stratigraphic rocks as a practice of identifying minerals with their characters. Maps showing distribution of occurrences of such stratigraphic rocks and land/sea distribution will also be practiced.				

Course Title	ESMJ67: Field Geology				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	---	02	02	
	Hour of Teaching (Total)	---	60	60	
Course Objectives (not more than 100 words)	The course focuses on advanced level of geological field training in a suitable area within the country.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	A field visit will be conducted to selected outstation locations, designed to provide hands-on geological field training. This training will focus on various aspects of geology, with an emphasis on applying the knowledge acquired from different branches of geology studied so far.				60
Texts / References					
Learning Outcomes (not more than 100 words)	The student will have first and experience the real exercises what professional geologists need to perform during their field visits under the guidance of their teachers.				

Course Title	ESMV61: Agriculture Geology				
Category of Course	VOC				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	02	02	04	
	Hour of Teaching (Total)	30	60	90	
Course Objectives (not more than 100 words)	Educate students about agriculture with knowledge of geology in cultivation. Develop operational resources for the efficient and cost effective implementation of agriculture education involving concept of mineralogy and geology.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Introduction to agrogeology, edaphology, micronutrients, agrominerals (apatite, carbonate, malachite, scoria, zeolite, clay), etc.				5
II	Soil Science and Pedology: Mineralogy and structure of soil, soil forming processes, soil classification and mapping, physical, chemical, biological and fertility properties of soil, pedogeochemistry, soil management and conservation.				10
III	Basic in Agronomy: Agroindustries, Agroecosystems, Agro-climatic Zones, Manures and Fertilizers, Weeds, Herbicides, Insecticides, Fungicides..				5
IV	Types of Soil, Distribution of Soil in India, Soil Taxonomy, Cropping and Farming System, Harvesting, Organic Agriculture, Crop Profiles, Soil and Water Conservation, Sustainable Agriculture.				10
V	Study of soil forming rocks and minerals. Determination of physical properties of soil. Study of soil profile in field. Study of soil map of India. Estimation of soil loss. Measurement of soil loss. Preparation of contour maps. Assessing entrepreneurial potential, problem solving ability, managerial skills and achievement motivation, exercise in creativity, time audit, preparation of business plan and proposal writing, visit to entrepreneurship development institute and entrepreneurs. Pollution case studies - Field work: A visit to village to understand the problems being encountered by the villagers / farmers; Listening and note taking, writing skills, oral presentation skills; field diary and lab record; indexing, footnote and bibliographic procedures.				30

VI	Identification of crops, seeds, fertilizers, pesticides and tillage implements; Studies on cropping pattern of different dry land areas in the country and demarcation of dry land area on map of India. Plotting and study of demand and supply curves and calculation of elasticities; Study of relationship between market arrivals and prices of some selected commodities. To get acquainted with university extension system. Group discussion - exercise; handling and use of audio-visual equipment's, digital camera and LCD projector; preparation and use of audio-visual aids, preparation of extension literature – leaflet, booklet, folder, pamphlet news stories and success stories; Presentation skills exercise; micro teaching exercise; Reading and comprehension of general and technical articles, precise writing, summarizing, abstracting; individual and group presentations.	30
Texts / References	1. S. C. Panda (2018) A Hand Book of Agriculture. AGROBIOS (INDIA), pp 1304. 2. S. Jothimani (2022) Text Book of Soil Science for Agriculture and Geology. Integrated Publications, New Delhi, pp 230. 3. S. C. Dutta (2022) Textbook on Physical Chemistry and Mineralogy of Soils. Indian Council of Agricultural Research, pp 360. 4. A. N. Puri (2015) Soil Science: An Elementary Text Book. New India Publishing Agency, pp 140. 5. H. D. Foth (1991) Fundamentals of Soil Science. 8th Edition, John Wiley & Sons Inc., pp 400. 6. Craig C. Sheaffer and Kristine M. Moncada (2016) Introduction to Agronomy Food, Crops, and Environment. Delmar Cengage Learning; 2nd edition, pp 608. 7. B. Chandrasekaran, K. Annadurai and E. Somasundaram (2010) A Text Book of Agronomy, pp 835. New Age International (P) Limited, Publishers, New Delhi. 8. Bipin B. Mishra (2020) The Soils of India. Springer Cham, pp 281.	
Learning Outcomes (not more than 100 words)	Students will reveal knowledge of the legal and ethical environment impacting agriculture organizations and exhibit an understanding geological concept. Students will be able to reveal critical thinking and problem-solving skills in the agricultural industry with the knowledge of geology as they apply to a variety of animal and or plant production systems.	

Semester: VII – H

Course Title	ESMJ71: Geomorphology and Geotectonics				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	03	---	03	
	Hour of Teaching (Total)	45	---	45	
Course Objectives (not more than 100 words)	The course consists of the significance of geomorphology, development of denudational landforms, different morphometric and morphotectonic indicators, geomorphology of India, and application of geomorphology in different branches of geology. Geotectonics course will help students understand the history of plate tectonics and how it shaped our planet earth and also it will deduce the knowledge of plate tectonics and use it as a tool for understanding mountain building activities on the earth.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Exploration of basic concepts and the significance of geomorphology. Study of the erosion cycle, fluvial landforms, and drainage patterns. Analysis of landform evolution in aeolian, marine, glacial, and karst environments. Introduction to morphometric analysis, morpho-chronology, and the role of neotectonics, including geomorphological indicators, active faults, drainage modifications, and recurrent seismic activity.				7
II	Examination of the geomorphological characteristics of the Peninsular, Extra-Peninsular regions, and the Indo-Gangetic plains. Discussion on the applications of geomorphology in mineral prospecting, civil engineering, military operations, hydrogeology, and environmental studies.				7
III	Overview of geotectonic principles, including the theories of continental drift, seafloor spreading, and convection currents. Study of paleomagnetism, polar wandering, and the reversal of Earth's magnetic field. Analysis of the geomagnetic timescale and the key geotectonic features of oceans, continents, and continental margins.				7
IV	Detailed study of plate tectonics, including the principles governing plate movements, force balance, and mantle plume models. Examination of orogeny and epeirogeny, and the structural anatomy of orogenic belts.				7
V	Analysis of the geodynamic processes that have shaped the evolution of the Himalayan mountain range.				2
Texts References	Geomorphology: - Small, R. J. (1978): Study of Landforms: A Textbook of Geomorphology (2nd Edition), Cambridge University Press. - Burbank, D. W., & Anderson, R. S. (2011): Tectonic Geomorphology, John Wiley & Sons. - Sharma, H. S. (1990): Indian Geomorphology, Concept Publishing Co., New Delhi. - Thornbury, W. D. (2004): Principles of Geomorphology (2nd Edition), CBS Publication. - Kale, V. S., and Avijit Gupta (2010): Introduction to Geomorphology, University Press. - Bloom, A. L. (2011): Geomorphology: A Systematic Analysis of Late Cenozoic Landforms (3rd Edition), Rawat Publications. Geotectonics: - Moore, Eldridge M., and Twiss, Robert J. (2015): Tectonics, W. H. Freeman. - Condie, Kent C. (1989): Plate Tectonics and Crustal Evolution (3rd Edition), Butterworth-Heinemann Ltd. - Windley, B. (1995): The Evolving Continents (3rd Edition), Wiley-Blackwell. - Davies, G. F. (1999): Dynamic Earth: Plates, Plumes and Mantle Convection, Cambridge University Press. - Keller, E. A., and Pinter, N. (2001): Active Tectonics (2nd Edition), Pearson Publications. - Kearey, P., Klepeis, K. A., and Vine, F. J. (2009): Global Tectonics (3rd Edition), Wiley-Blackwell.				

Learning Outcomes (not more than 100 words)	After completing the course, the students will be able to understand the significance and application of geomorphology in different geological prospecting. They will also be familiar with continental drift theory, sea floor spreading, and paleomagnetism, and they may gain a better understanding of plate tectonics and the evolution of the Himalayas.				
Course Title	ESMJ72: Foundations of Engineering Geology and Surveying				
Category of Course	Major				
Credits & Hour of Teaching		Theory	Practical	Cumulative	
	Credits	03	---	03	
	Hour of Teaching (Total)	45	---	45	
Course Objectives (not more than 100 words)	To study and identify different types natural materials like rocks & soil with reference to geohazards, construction and excavation. • To know the physico-mechanical properties of rocks and rockmass • To know the importance of testing and geotechnical considerations for slopes, tunnel and dam construction. To attain the knowledge about the basics of surveying and theatrical background.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Geotechnical properties of rock and soil, aggregates and ground investigation methods, Rockmass description and classification schemes, types of strength- uniaxial compressive strength, tensile strength, shear strength, laboratory tests.				15
II	Engineering geological considerations for dam, roads and tunnels. Ground subsidence, slope stability & rock excavation. Fundamental Concepts and principles; Primary division and classification of surveys; Common methods of surveying: Reconnaissance survey, Offsetting, Radiation, and triangulation; Open and close traversing; Leveling. Accuracy and errors in surveying; Basic elements of map preparation and map reading;				15
III	Working principles and use of different Surveying Instruments: Chain, Plane Table, Prismatic Compass, Dumpy Level, Theodolite and Total Station. Concept of Global Navigation Satellite Systems (GNSS): GPS and Navic.				15
Texts / References	Foundations of Engineering Geology: 1. S. Gangopadhyay (2012): Engineering Geology, Oxford Press. 2. T. Waltham (2009): Foundations of Engineering Geology (Third Edition), Taylor & Francis. 3. D. G. Price (2009): Engineering Geology - Principles & Practice, Springer. Surveying: 1. Bernhard Hofmann-Wellenhof, Herbert Lichtenegger, and James Collins (2012): Global Positioning System: Theory and Practice (4th Edition), Springer Science & Business Media. 2. Punmia, B. C., Jain, A. K., and Jain, A. K. (2016): Surveying, Vol. 1 & Vol. 2, Laxmi Publications (P) Limited, New Delhi. 3. Ray Chowdhary, K. P. (1987): Surveying (Plane and Geodetic), Oxford & IBH Pub. Co., New Delhi. 4. Shahani, P. B. (1978): Textbook of Surveying, Vol. I, Oxford & IBH Pub. Co., New Delhi. 5. Venkataramaiah, C. (2011): Textbook of Surveying, Orient Blackswan Private Limited, New Delhi.				
Learning Outcomes (not more than 100 words)	The students will get the basic knowledge about natural material vis-a-vis construction and excavation. • The students will understand the influence of natural processes and geological factors on civil structures and help them to take decision while planning, design and execution stage of the structures in their professional life. • The students will know the significance of geological investigations for civil engineering projects and site selection as well as engineering geological testing. The student will be able to comprehend the fundamentals of surveying and their uses in geological sciences.				

Course Title	ESMJ73: Geochemistry and Instrumentation				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	03	---	03	
	Hour of Teaching (Total)	45	---	45	
Course Objectives (not more than 100 words)	The objective of the course is to equip students with a deep understanding of Earth's chemical composition and processes, as well as the modern analytical techniques used to study them. Students will learn to analyze geochemical data, understand geochemical cycles and isotope systems, and apply thermodynamic principles to geological processes. The course aims to develop proficiency in using advanced geochemical instruments and techniques, ensuring students can conduct precise and accurate geochemical analyses. Ultimately, students will be prepared to apply their knowledge to address fundamental problems in geoscience using geochemistry as a proxy.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Geochemistry: Earth System Science and various reservoirs. Introduction to the chemical composition and properties of the atmosphere, lithosphere, hydrosphere, and biosphere. Geochemical cycles. Concept of biogeochemical cycle. Concept of equilibrium. Entropy, enthalpy, Gibbs free energy, and laws of thermodynamics and their applications in geosystems. Meteorites, their classification, mineralogy, and origin. Geochemical classification of elements. Element partitioning in mineral/rock formation and the concept of distribution coefficient. Utility of trace elements in the petrogenesis of rocks. Importance of REE (Rare Earth Elements) in petrogenetic interpretations. Interpretation of REE patterns. Mineral formula calculations and their implications in understanding the physico-chemical conditions of magmas. Sampling procedures and introduction to state-of-the-art instruments & important analytical techniques used in geochemistry and their applications				15
II	Importance of isotopic study in the field of earthscience, Stable isotope geochemistry of carbon, oxygen, and sulphur and its applications. Non-traditional isotopes and their applications. Radiogenic isotopes. Decay scheme of K-Ar, U-Pb, Rb-Sr and Sm-Nd. Radiometric dating of single minerals and whole rocks. Petrogenetic implications of Sm-Nd, Rb-Sr, Re-Os and Pb-Pb systems. C^{14} dating and its application in geosciences. Introduction to sedimentary geochemistry. Geochemical processes involved in rock weathering and soil formation. Mineral stability in Eh-Ph diagrams.				15
III	Instrumentation An overview of instrumentation techniques in Geosciences. Their role and importance in addressing fundamental and applied aspects. Sample preparation techniques for various instruments. Precision, accuracy, calibration, and standards. Destructive and non-destructive techniques. Principle and application of the following instrumental techniques in Mineral Sciences: • Microscopes • UV Lamp • X-ray Diffractometer • Image Analyzer • Electron Probe Micro Analyzer • Scanning Electron Microscope • Infrared Spectrometer • Raman Spectroscope • FTIR (Fourier Transform Infrared Spectroscopy) Principle and application of the following instrumental techniques in Geochemistry:				15

	• Atomic Absorption Spectrometer • X-ray Fluorescence Spectrometer • X-ray Photoelectron Spectroscopy • Inductively Coupled Plasma Analyzers • Isotope Ratio Mass Spectrometer • Thermal Ionization Mass Spectrometer • Laser-Based Instruments Laboratory accreditation, maintenance, and quality control.	
Texts / References	Geochemistry: 1. Robin Gill (2015): Chemical Fundamentals of Geology and Environmental Geoscience, John Wiley & Sons Ltd. 2. Alan P. Dickins (2005): Radiogenic Isotope Geology, Cambridge University Press. 3. Hoefs, J. (1980): Stable Isotope Geochemistry, Springer and Verlag. 4. Hugh R. Rollinson (2007): Early Earth Systems: A Geochemical Approach, Blackwell Publishing Ltd. 5. Gunter Faure (1977): Principles of Isotope Geology, John Wiley & Sons Ltd. 6. Hugh R. Rollinson and V. Pease (2022): Using Geochemical Data: Evaluation, Presentation and Interpretation, Pearson Prentice Hall. 7. Potts, P. J. (1992): A Handbook of Silicate Rock Analysis, Springer. 8. Albarde Francis (2003): Geochemistry - Introduction, Cambridge University Press. 9. Kula C. Misra (2012): Introduction to Geochemistry: Principles and Applications, Wiley-Blackwell. Instrumentation 1. Dhanaraju, R. (2009): Handbook of Geochemistry: Techniques and Applications in Mineral Exploration, Geological Society of India. 2. Perkins, D. (2013): Mineralogy, Prentice Hall. 3. Reed, S. J. B. (2006): Electron Microprobe Analysis and Scanning Electron Microscopy in Geology, Cambridge University Press. 4. J. Carey (2016): Geoscience: Instrumentation and Analytical Techniques.	
Learning Outcomes (not more than 100 words)	By completing the course, students will achieve a thorough understanding of the chemical composition and properties of Earth's atmosphere, hydrosphere, lithosphere, and biosphere. They will learn to analyze the distribution and behavior of chemical elements in rocks and minerals. Students will gain skill in knowing the basic fundamentals of modern geochemical instrumentation and techniques. They will develop knowledge in interpreting geochemical data, understanding isotopic systems, and applying geochemical principles to solve geological problems. Additionally, students will be equipped to conduct independent research and contribute to advancements in geosciences through the use of geochemical tools and methodologies.	

Course Title		ESMJ74: Exploration Geosciences			
Category of Course		Major			
Credits& Hour of Teaching			Theory	Practical	Cumulative
		Credits	03	---	03
		Hour of Teaching (Total)	45	---	45
Course Objectives (not more than 100 words)		This course educates students on the principles and practices for locating and evaluating mineral and petroleum resources. Key objectives include understanding geological methods for mineral exploration, criteria for effective mineral search, and stages of exploration. The course covers geochemical exploration techniques, including dispersion mechanisms and mapping. Additionally, it introduces geophysical methods such as gravity, magnetic, electrical, and seismic techniques for subsurface imaging. Students also learn about well-logging methods, porosity, permeability, and drilling techniques relevant to exploration. Overall, the course prepares students to apply interdisciplinary approaches in geological and geophysical data interpretation to assess and delineate mineral and petroleum prospects effectively.			
Course Content		The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content				Hr. of Teaching
I	Principles of geological methods for mineral exploration, including criteria and guidelines for mineral prospecting. Exploration stages: an overview of geochemical exploration, its methodologies, and the significance of dispersion; geochemical mapping. Introduction to geobotanical exploration techniques				15
II	Core principles of geological methods for petroleum exploration, including identification and characterization of petroleum source rocks. Analysis of organic matter's quantity, type, and maturation, along with the fundamentals of petroleum generation and migration. Evaluation of oil and gas prospects using geochemical modeling.				10
III	Overview of geophysics, including fundamental theories of gravity and magnetic methods, as well as the densities and magnetic susceptibilities of rocks and minerals. Examination of the electrical properties of rocks and minerals, self-potential, D.C. resistivity, and induced polarization (IP) methods. Introduction to electromagnetic (EM) induction, fundamentals of seismic theory, reflection and refraction methods. Understanding the principles of radioactivity and the radioactivity of rocks and minerals				10
IV	Well-logging and drilling: Understanding the borehole environment and key concepts such as porosity, permeability, and saturation. Overview of well logging methods and their interpretation. Exploration drilling: objectives, types, and advantages.				10
Texts / References	1. Moon, C. J., Michael, K. G., Whateley, and Evans, A. M. (2006): Introduction to Mineral Exploration, Blackwell Publishing House. 2. Banerjee, P. K., and Ghosh, S. (1997): Elements of Prospecting for Non-Fuel Mineral Deposits. 3. Tissot, B. P., and Welte, D. H. (1984): Petroleum Formation and Occurrence, Springer-Verlag. 4. North, F. K. (1985): Petroleum Geology, Allen &Unwin. 5. Selley, R. C. (1998): Elements of Petroleum Geology, Academic Press. 6. Schlumberger: Log Interpretation Principles/Application, Schlumberger Wireline& Testing, 225 Schlumberger Drive, Sugar Land, Texas 77478. 7. Musset, A. E., and Khan, M. A. (2000): An Introduction to Geological Geophysics, Cambridge University Press. 8. Lille, R. (1998): Whole Earth Geophysics: An Introductory Textbook for Geologists and Geophysicists, Pearson; 1st edition.				
Learning Outcomes (not more than 100 words)	Upon completing Exploration Geosciences, students will: 1. Apply fundamental geological methods to identify and assess mineral and petroleum resources. 2. Analyze criteria and techniques for effective mineral exploration and petroleum prospecting. 3. Demonstrate proficiency in geochemical and geophysical exploration methods, including interpreting data from gravity, magnetic, electrical, and seismic surveys. 4. Evaluate the geological factors influencing resource distribution and quality. 5. Utilize well-logging techniques to interpret subsurface characteristics such as porosity, permeability, and fluid saturation.				

Course Title	ESMJ75: Fuel Geology				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	03	---	03	
	Hour of Teaching (Total)	45	---	45	
Course Objectives (not more than 100 words)	To learn about the important sources of energy and laying conceptual foundation about their origin, formation and deposits. Understanding the geological framework of these natural exhaustible resources.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Coal Geology: Definition and origin of coal. Chemical characteristics of coal. Coal petrology – concept and classification of ‘Coallithotypes’ and ‘maceral’. Techniques and methods of coal microscopy. Elementary knowledge of the application of reflectance and fluorescence microscopy.				10
II	Application of coal petrology: Classification of coal in terms of rank, grade, and type. Classification for coking and non-coking coals. Elementary idea about coal preparation.				10
III	Elementary idea about Coal Bed Methane and Underground Coal Gasification. Geological andgeographical distribution of coal and lignite deposits in India.Coal exploration and estimation of coal reserves. Indian coal reserves and production of coal in India.				10
IV	Petroleum Geology:Petroleum – its composition. Origin (Formation of source rocks-kerogen,organic maturation and thermal cracking of kerogen) and migration of petroleum.Reservoirrocks-porosityandpermeability.Reservoirtraps–structural, stratigraphic and combinationtraps.Oilfieldfluids–water, oil and gas. On shore and offshore petroliferous basins of India.				15
Texts / References	1. Isabel Suárez-Ruiz, John Crelling. 2008. Applied Coal Petrology: The Role of Petrology in Coal Utilization, Academic Press. 2. Taylor, G.H., Teichmuller, M., Davis, A., Diessel, C.F.K., Littke, R., and Robert P. 1998. Organic Petrology, GebruderBorntraeger, Stuttgart. 3. Chandra, D., Singh, R.M., Singh, M.P. 2000. Textbook of Coal (Indian context), Tara Book Agency, Varanasi. 4. Singh, M.P. (Ed.) 1998. Coal and Organic Petrology, Hindustan Publishing Corporation, New Delhi. 5. Scott, A.C. 1987. Coal and Coal-bearing Strata: Recent Advances, The Geological Society of London, Publication no. 32, Blackwell Scientific Publications. 6. Stach, E., Mackowsky, M.-Th., Taylor, G.H., Chandra, D., Teichmuller, M., and Teichmuller, R. 1982. Stach Textbook of Coal Petrology, GebruderBorntraeger, Stuttgart. 7. Holson, G.D. and Tiratso, E.N. 1985. Introduction to Petroleum Geology, Gulf Publishing, Houston, Texas.Tissot, B.P. and Welte, D.H. 1984. Petroleum Formation and Occurrence, Springer-Verlag. 8. North, F.K. 1985. Petroleum Geology, Allen &Unwin.				
Learning Outcomes (not more than 100 words)	Coal and Petroleum as a natural resource in the geological realm.				

Course Title	ESMJ76: Practical connected with ESMJ71				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	---	01	01	
	Hour of Teaching (Total)	---	30	30	
Course Objectives (not more than 100 words)	The course ES-CC-411 includes practical exercises on drainage extraction, hypsometric analysis, and geomorphological study using topographic sheets, aerial photos, and satellite imagery. It also encompasses the study of different geotectonic maps of the major plates of the Earth.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Geomorphology: Drainage and Slope Morphometry, Hypsometry				6
II	Geomorphology through toposheets				6
III	Geomorphology through aerial photos and satellite Imagery, and Terrain aspect mapping.				6
IV	Geotectonics: Analysis of geotectonic maps focusing on the major lithospheric plates and their movements. Examination of key geotectonic features such as mid-oceanic ridges, aseismic ridges, island arcs, trenches, subduction zones, fracture zones, hot spots, and triple junctions.				6
V	Exploration of the characteristics and development of contemporary continental margins and orogenic belts. Detailed study of plate tectonic maps across different geological time periods, focusing on the evolution and interaction of tectonic plates throughout Earth's history.				6
Texts / References	Geomorphology: - Small, R.J. 1978. Study of Landforms: A Textbook of Geomorphology (2nd Edition), Cambridge University Press. - Halis, J.R. 1983. Applied Geomorphology. - Thornbury, W.D. 2004. Principles of Geomorphology. 2nd Edition, CBS Publication. - Kale, V.S. and Avijit Gupta 2010. Introduction to Geomorphology, University Press. - Bloom, A.L. 2011. Geomorphology: A Systematic Analysis of Late Cenozoic Landforms. 3rd Edition, Rawat Publications. - Summerfield, M.A. 2011. Geomorphology and Global Tectonics, Wiley India Pvt Ltd. Geotectonics: - Gass, I.G. et al. 1982. Understanding the Earth, Artemis Press (Pvt) Ltd., U.K. - Condie, Kent C. 1989. Plate Tectonics and Crustal Evolution. 3rd Edition, Butterworth-Heinemann Ltd. - Windley, B. 1995. The Evolving Continents. 3rd Edition, Wiley-Blackwell. - Davies, G.F. 1999. Dynamic Earth: Plates, Plumes and Mantle Convection, Cambridge University Press. - Keller, E.A. and Pinter, N. 2001. Active Tectonics. 2nd Edition, Pearson Publications. - Kearey, P., Klepeis, K.A. and Vine, F.J. 2009. Global Tectonics. 3rd Edition, Wiley-Blackwell.				
Learning Outcomes (not more than 100 words)	After completing the practical exercises in course ES-CC-411, students will gain a better understanding of subsurface geology, including lithology and deformation based on surface manifestations. They will also acquire knowledge about different plate boundaries and their interrelationship on Earth.				

Course Title	ESMJ77: Practical connected with ESMJ72				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	---	01	01	
	Hour of Teaching (Total)	---		30	
Course Objectives (not more than 100 words)	To study and identify different types natural materials like rocks & soil with reference to geohazards, construction and excavation. - To know the physico-mechanical properties of rocks and rockmass - To know the importance of testing and geotechnical considerations for slopes, tunnel and dam construction. - To have field knowledge about the basics of surveying and theatrical background.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Engineering Geology: Testing for UCS, Testing for Point Load Strength, Testing for Tensile Strength, Kinematic Analysis for Slope Instability, Slake Durability Index, Analytical Solutions for Stability Analysis, Graphical Solutions for Slope Instability, Sieve Analysis, Python Programming for Geo-engineering Applications.				15
II	Surveying: Laboratory and field exercises by surveying instruments viz. Chain, Prismatic Compass, Plane Table, Dumpy Level, Theodolite, and Total Station: Survey of a piece of land by means of common methods of surveying using the suitable surveying instruments.				15
Texts / References	Foundations of Engineering Geology S. Gangopadhyay, 2012, Engineering Geology, Oxford Press. T. Waltham, 2009, Foundations of Engineering Geology (Third edition), Taylor & Francis. D. G. Price, 2009, Engineering Geology - Principles & Practice, Springer. Surveying - Bernhard Hofmann-Wellenhof, Herbert Lichtenegger, James Collins, 2012: Global Positioning System: Theory and Practice, 4th edition, Springer Science & Business Media. - Punmia, B.C., Jain, A.K., and Jain, A.K., (2016): Surveying, Vol. 1 & Vol. 2, Laxmi Publications (P) Limited, New Delhi. - Ray Chowdhary, K.P., (1987): Surveying (Plane and Geodetic), Oxford & IBH Pub. Co., New Delhi. - Shahani, P.B., (1978): Textbook of Surveying, Vol. I, Oxford & IBH Pub. Co., New Delhi. - Venkataramaiah, C. (2011): Textbook of Surveying, Orient Blackswan Private Limited, New Delhi.				
Learning Outcomes (not more than 100 words)	The students will get the basic knowledge about natural material vis-avis construction and excavation. - The students will understand the influence of natural processes and geological factors on civil structures and help them to take decision while planning, design and execution stage of the structures in their professional life. - The students will know the significance of geological investigations for civil engineering projects and site selection as well as engineering geological testing. The student will be able to comprehend the fundamentals of surveying and their uses in geological sciences.				

Course Title	ESMJ78: Practical connected with ESMJ73			
Category of Course	Major			
Credits& Hour of Teaching		Theory	Practical	Cumulative
	Credits	---	01	01
	Hour of Teaching (Total)	---	30	30
Course Objectives (not more than 100 words)	The objective of the paper is to provide students with hands-on experience in analyzing and interpreting rock samples using geochemistry as a proxy. Students will learn to perform rock analyses, calculate mineral formulas, and prepare and interpret geochemical diagrams and models. The course aims to develop practical skills in data plotting and interpretation, including rare earth element (REE) analysis, and to apply geochemical modeling to understand geological processes. Ultimately, students will be equipped with the practical expertise needed to handle geochemical data and conduct advanced research in geosciences.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Rock analyses (Rapid method of silicate analysis); Mineral formula calculations; Preparation of classificatory and variation diagrams and their interpretation; Plotting of REE data and their interpretation; Geochemical modeling.			30
Texts / References	1. Robin Gill (2015) Chemical Fundamentals of Geology and Environmental Geoscience, John Wiley & Sons Ltd. 2. Alan P. Dickins (2005) Radiogenic Isotope Geology, Cambridge University Press. 3. Hoefs, J. (1980) Stable Isotope Geochemistry, Springer and Verlag. 4. Hugh R. Rollinson (2007) Early Earth Systems: A Geochemical Approach, Blackwell Publishing Ltd. 5. Gunter Faure (1977) Principles of Isotope Geology, John Wiley & Sons Ltd. 6. Hugh R. Rollinson and V. Pease (2022) Using Geochemical Data: Evaluation, Presentation and Interpretation, Pearson Prentice Hall. 7. Potts, P.J. (1992) A Handbook of Silicate Rock Analysis, Springer. 8. Albarde Francis (2003) Geochemistry - Introduction, Cambridge University Press. 9. Kula C. Misra (2012) Introduction to Geochemistry: Principles and Applications, Wiley-Blackwell. 10. Instrumentation in Geosciences 11. Dhanaraju, R. (2009) Handbook of Geochemistry: Techniques and Applications in Mineral Exploration, Geological Society of India. 12. Perkins, D. (2013) Mineralogy, Prentice Hall. 13. Reed, S.J.B. (2006) Electron Microprobe Analysis and Scanning Electron Microscopy in Geology, Cambridge University Press. 14. J. Carey (2016) Geoscience: Instrumentation and Analytical Techniques.			
Learning Outcomes (not more than 100 words)	Upon completing the paper, students will gain ability in analyzing rocks using rapid silicate analysis methods. They will be skilled in calculating mineral formulas from EPMA data, preparing and interpreting classificatory and variation diagrams, and plotting and interpreting rare earth element (REE) data. Students will develop the ability to perform geochemical modeling to understand and predict igneous processes. This practical experience will enhance their capability to apply geochemical techniques to geological problems, interpret complex geochemical data, and contribute to advanced geochemical research and analysis.			

Course Title	ESMJ79: Practical connected with ESMJ74				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	02	---	02	
	Hour of Teaching (Total)	30	---	30	
Course Objectives (not more than 100 words)	This course gives hands-on training in using geological and geophysical instruments for mineral and petroleum exploration. Practicing geochemical analysis techniques to interpret mineralization indicators. Conducting geophysical data acquisition and processing for subsurface imaging. Interpreting well-logging data to evaluate reservoir characteristics such as porosity and permeability. Enhancing skills in geological mapping and integrating data sets to identify exploration targets. Collaborating effectively in teams to solve practical problems and present findings through comprehensive laboratory reports.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Development of geological cross sections using borehole data; delineating stripping boundaries on geological cross sections and calculating ore reserves. Creation of geological maps and sections to infer geological history about petroleum prospects, along with oil reserve estimation. Interpret geophysical maps and computational techniques for analyzing and interpreting well-log data.				30
Texts / References	1. Moon, C.J., Micheal, K.G., Whateley, and Evans, A.M. (2006) Introduction to Mineral Exploration, Blackwell Publishing House. 2. Banerjee, P.K. and Ghosh, S. (1997) Elements of Prospecting for Non-Fuel Mineral Deposits. 3. Tissot, B.P. and Welte, D.H. (1984) Petroleum Formation and Occurrence, Springer-Verlag. 4. North, F.K. (1985) Petroleum Geology, Allen &Unwin. 5. Selley, R.C. (1998) Elements of Petroleum Geology, Academic Press. 6. Schlumberger Log Interpretation Principles/Application, Schlumberger Wireline& Testing 225, Schlumberger Drive Sugarland, Texas 77478. 7. Musset, A.E. and Khan, M.A. (2000) An Introduction to Geological Geophysics, Cambridge University Press. 8. Lille, R. (1998) Whole Earth Geophysics: An Introductory Textbook for Geologists and Geophysicists, Pearson; 1st edition.				
Learning Outcomes (not more than 100 words)	Upon completing the Exploration Geosciences Practical course, students will: 1. Demonstrate proficiency in using geological and geophysical instruments for practical mineral and petroleum exploration. 2. Apply geochemical analysis techniques to interpret indicators of mineralization. 3. Conduct geophysical data acquisition and processing to interpret subsurface structures and resources. 4. Interpret well-logging data to evaluate reservoir properties such as porosity and permeability. 5. Develop skills in geological mapping and integrating diverse datasets to identify exploration targets accurately. Collaborate effectively in teams to solve practical challenges and communicate findings through detailed laboratory reports, preparing them for professional roles in exploration geoscience.				

Course Title	ESMJ710: Practical connected with ESMJ75				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	---	01	01	
	Hour of Teaching (Total)	---	30	30	
Course Objectives (not more than 100 words)	To learn about the important sources of energy and laying conceptual foundation about their origin, formation and deposits. Understanding the geological framework of these natural exhaustible resources.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Macroscopic characterization of banded coals.Completion of outcrop in the given maps and calculation of coal reserve.Preparation of polished particulate mounts of coal.Microscopic examination of polished particulate mounts (identification of macerals).				15
II	Macroscopic and microscopic study of cores and well cuttings.Study of geological maps and sections of important oil fields of India				15
Texts / References	1. Isabel Suárez-Ruiz, John Crelling. (2008) Applied Coal Petrology: The Role of Petrology in Coal Utilization, Academic Press. 2. Taylor, G.H., Teichmuller, M., Davis, A., Diessel, C.F.K., Littke, R., and Robert, P. (1998) Organic Petrology, Gebruder Borntraeger, Stuttgart. 3. Chandra, D., Singh, R.M., Singh, M.P. (2000) Textbook of Coal (Indian Context), Tara Book Agency, Varanasi. 4. Singh, M.P. (Ed.) (1998) Coal and Organic Petrology, Hindustan Publishing Corporation, New Delhi. 5. Scott, A.C. (1987) Coal and Coal-Bearing Strata: Recent Advances, The Geological Society of London, Publication No. 32, Blackwell Scientific Publications. 6. Stach, E., Mackowsky, M.-Th., Taylor, G.H., Chandra, D., Teichmuller, M., and Teichmuller, R. (1982) Stach Textbook of Coal Petrology, Gebruder Borntraeger, Stuttgart. 7. Holson, G.D. and Tiratso, E.N. (1985) Introduction to Petroleum Geology, Gulf Publishing, Houston, Texas. 8. Selley, R.C. (1998) Elements of Petroleum Geology, Academic Press.				
Learning Outcomes (not more than 100 words)	Coal and Petroleum as a natural resource in the geological realm.				

Course Title	ESMJ711: Advanced Geological Field Training				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	---	04	04	
	Hour of Teaching (Total)	---	120	120	
Course Objectives (not more than 100 words)	The students will travel to a region with interesting and accessible geology and use traditional and modern tools to understand the geology of the region. Students will then produce a field report, including analysis of various geological data and relevant cross-sections. The course aims to familiarize students with the real-world conditions and challenges faced by professional geoscientists. Additionally, students will visit scientific organizations, mining sites, institutes, and companies of national repute to gain insights into the professional working environment and operational practices in the field of geosciences.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
	Comprehensive field training will be provided at selected outstation locations within the country. Students are also expected to visit prominent scientific organizations, mining sites, dams/tunnels/excavations and institutes to gain familiarity with the practical fieldwork and operational environments of professional geoscientists.				120
Texts / References					
Learning Outcomes (not more than 100 words)	1. Students will gain practical knowledge and skills through direct involvement in geological fieldwork conducted at various outstation locations within the country. It will bridge the gap between classroom learning and field application, enabling students to apply theoretical concepts in real-world scenarios. 2. Students will have the chance to interact with professionals and experts in the field, providing opportunities for networking and future career development. They will know the actual conditions and challenges encountered by professional geoscientists in the field, enhancing their readiness for future professional roles. 3. By visiting scientific organizations, mining sites, institutes, and companies of national repute, students will gain valuable insights into the operational practices and professional working environments within the geosciences sector. 4. It will help students develop critical thinking and problem-solving skills pertinent to geological challenges.				

Course Title	Same as major ESMJ71 and ESMJ75				
Category of Course	Minor				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	4	---	04	
	Hour of Teaching (Total)	60	---	60	

Semester VIII- H

Course Title	ESMJ81: Marine Geology, Oceanography and Climate				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	03	---	03	
	Hour of Teaching (Total)	45	---	45	
Course Objectives (not more than 100 words)	The course introduces the origin and evolution of oceans, focusing on their physical, chemical, and biological changes throughout the geological past. It covers various micropaleontological techniques to reconstruct past climates and oceanic conditions. The course also explores various oceanic processes and their impacts on climate and the environment. Additionally, it addresses the application of these studies to understand the dynamics of monsoons and Global warming, providing an overview of climate policies and Global agreements.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	An Introduction to Marine Geology: Origin and morphology of the ocean floor, ocean margins, and coastal features. Marine sediments, their types, deposition, distribution, and diagenetic changes. Mineral resources.				15
II	Micropaleontology: Definition and scope, microfossil types, analysis techniques, and applications in paleoceanographic/paleoclimatic reconstructions.				15
III	Physical and chemical properties of seawater, ocean currents and circulations; fundamentals of climatology: Ocean's role in climate change, Global warming and greenhouse gases; an overview of climate policy and Global agreements like the Kyoto Protocol and Paris Agreement.				15
Texts / References	1. Pratul Kumar Saraswati and M.S. Srinivasan, 2016. <i>Micropaleontology: Principles and Applications</i>. Springer Cham Heidelberg New York Dordrecht London. 2. Alan P.T. and Harold V.T., 2019. <i>Essentials of Oceanography</i>. Pearson Education. 3. Wright, J.M. and Colling, A., 2013. <i>Seawater: Its Composition, Properties and Behaviour: Prepared by an Open University Course Team</i>. Elsevier. 4. Kennett, James P., 1982. <i>Marine Geology</i>. Prentice-Hall, Englewood Cliffs, N.J. 5. Rohli, R.V. and Vega, A.J., 2017. <i>Climatology</i>. Jones & Bartlett Learning. 6. John L. Mero, 1964. <i>Mineral Resources of the Sea</i>. Elsevier Publishing Company, University of Virginia.				
Learning Outcomes (not more than 100 words)	By the end of this course, students will be able to understand the role of oceans in regulating climate and sustaining life on Earth. They will be proficient in using micropaleontological techniques to reconstruct past climates and oceanic conditions, gaining insights into the mechanisms behind various climatic and oceanographic processes. Additionally, they will understand the dynamics of monsoons and Global warming and be familiar with relevant climate policies and Global agreements.				

Course Title	ESMJ82: Georesource and Mineral Policy				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	03	---	03	
	Hour of Teaching (Total)	45	---	45	
Course Objectives (not more than 100 words)	This course is an introductory course for Georesource. Promote best practices for efficient and sustainable mineral production, use, recycling, and environmental rehabilitation Provide an overview of the legal and policy framework on the mining sectors at national and international framework. To understand the causal forces which have been transforming the laws and policies on the mineral industries.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Concepts of Georesources, Classification of Georesources, Renewable and Non-Renewable Resources, Mineral, Energy, Water, and Land Resources. Greenfield and Brownfield Georesources.				10
II	Classification of Mineral Resources, Mineral Resource Exploration, Feasibility and Production, Preservation of Environment and Sustainable Development of Mineral Resources, Mineral Conservation, World Mineral Resources and Reserves. Geopolitics on Mineral Resources, Blue Economy, Marine Mineral Resources.				10
III	Concept of Mineral Economics, Mineral Industry in India, Definition of Mineral and Mineral Deposit in Commercial and Legal Literature. Mineral Laws and Taxation, Mineral Market, Internal and External Trade, Demand and Supply, Consumption and Substitution.				10
IV	National Mineral Policy and Legislation, Mines and Minerals (Development & Regulation) Act in India, Law of Sea (UNCLOS), International Seabed Authority (ISA), Exclusive economic zones (EEZ), Legal continental shelf (LCS).				15
Texts / References	1. K.L. Shrivastava and Arun Kumar (2014) Geo-Resources. Scientific Publishers, pp 672. 2. J. Craig, D. Vaughan and B. Skinner (2010) Earth Resources and the Environment. Pearson Publication, pp 528. 3. S.E. Kesler and A.C. Simon (2010) Mineral Resources, Economics and the Environment. Cambridge University Press, pp 392. 4. K.S. Valdiya (2016) Environmental Geology. Second Edition, McGraw Hill Education India Pvt. Ltd. 5. R.K. Sinha and N.L. Sharma (2019) Mineral Economics. 4th Edition, Oxford & IBH Publishing, pp 410. 6. K.N. Jetli (2011) Mineral Resources and Policy in India. New Century Publications, pp 356. 7. Fillmore C.F. Earney (2002) Marine Mineral Resources. Taylor & Francis, pp 405. 8. Pradeep Kumar Jain (2021) Mineral Policy, Mining Laws and Development, 2nd Edition, Scientific Publishers, pp 290. 9. P.L. Billon (2005) Geopolitics of Resource Wars: Resource Dependence, Governance and Violence. Frank Cass Publishers, pp 277. 10. Indian Mineral Year Book, Indian Bureau of Mines. 11. M.S. Cato (2009) Green Economics. Earthscan Publishing, London, pp 224. 12. A.M. Hussen (2000) Principles of Environmental Economics. Routledge London, pp 354.				
Learning Outcomes (not more than 100 words)	Demonstrate proficiency in practical skills used in the evaluation of mineral resources. Be able to apply knowledge and skills within all study specializations in a social economics. Use and evaluate the appropriate strategies and methods for exploration of natural resources. Apply the scientific approaches being applied to exploration and production from natural resources. Interpret and analyze the geological characteristics of the mineral deposits to develop and planning and formulate national mineral policy.				

Course Title	ESMJ83: Ground Water Hydrology and Soil Geology				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	02	---	02	
	Hour of Teaching (Total)	30	---	30	
Course Objectives (not more than 100 words)	The course focuses on teaching basic concepts and principles of soil science and it's association with groundwater hydrology. Course covers the fundamentals soil genesis, soil types, it's mechanical properties, subsurface flow and transport, relation of groundwater flow to soil, introductory groundwater chemistry, and the management of contaminated groundwater. The objective is to train students through case studies and fieldwork, which help to develop critical thinking and problem-solving abilities essential for addressing contemporary soil and groundwater-related challenges.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Introduction and basic concepts of hydrogeology: Scope of hydrogeology and its societal relevance, Hydrologic cycle, water budget, Hydrological Components: Precipitation, Evapotranspiration, run-off, infiltration and subsurface movement of water. Origin of water on Earth, Vertical distribution of groundwater, Types of aquifers, Hydrogeological properties of water-bearing rock – porosity, permeability, transmissivity, storage coefficient, specific yield and specific retention, Groundwater level and its fluctuations, springs.				07
II	Groundwater flow, Darcy's law and its validity, intrinsic permeability and hydraulic conductivity, Basic concepts of Well hydraulics and Groundwater exploration, Steady flow in confined and unconfined aquifers. Groundwater. Physical and chemical properties of groundwater and its quality, Graphic representation of chemical data, Environmental isotopes and its application in hydrogeological studies. Sea water intrusion in coastal aquifers. Artificial recharge of groundwater, Surface and subsurface water interaction, Groundwater Provinces of India				08
III	Soil Geology: Concept of soil, components of soil, Processes of soil formation; factors affecting soil formation; soil profile; Phase relationship and grain size analysis; Classifications of soil, mineral and chemical composition of soils, Soil organic matter form and function; mineral stability during weathering; A brief introduction to methods of soil conservation. Concepts of soil structure and fabric: Soil fabric, soil structure, soil texture and field grading units; Peds and pedality, size and shape of peds, pedality, primary, secondary and tertiary structures and their interpretation; Voids-concepts, size, shape, arrangement and morphological classification.				08
IV	Paleosols - Field recognition, description, origin and causes; paleosol in stratigraphic records; significance of paleosols; paleosols and human evolution. Calcrete - Definition, classification, calcrete formation, pedogenic calcrete profiles, macrofeatures in calcretes, micromorphology (petrography), calcretes from Quaternary and ancient sedimentary sequences; significance of calcretes. Laterite - Characteristics, genesis, Indian occurrences.				07
Texts / References	Groundwater Hydrology: - Todd, D.K. (2006). Groundwater Hydrology, 2nd Ed., John Wiley & Sons, N.Y. - Davis, S.N., and DeWeist, R.J.M. (1966). Hydrogeology, John Wiley & Sons Inc., N.Y. - Karanth, K.R. (1987). Groundwater: Assessment, Development and Management, Tata McGraw-Hill Pub. Co. Ltd. Soil Geology: - Braddy, N.C. Nature and Properties of Soils. - Gerrard, A.J.J. Soil and Landforms. - GovindaRajan, S.V., & GopalaRao, K.H.G. Studies of Soils of India.				

	4. Gurrison, S. (1989). The Chemistry of Soils, Oxford University Press. 5. Hunt, C.B. Geology of Soils. 6. Jeffe, J.S. The A.B.C. of Soils. 7. Terzaghi, K., & Pock, R.G. Soil Mechanics in Engineering. 8. Taylor, D.W. Fundamentals of Soil Mechanics. 9. Wright, V. Paul (Editor). Paleosols: Their Recognition and Interpretation, Blackwell Scientific Publication. 10. Wright, V. Paul, and Tucker, M.E. (1991). Calcretes, Blackwell Scientific Publication.			
Learning Outcomes (not more than 100 words)	1. Analyze hydrogeological systems, including groundwater quality and the causes of groundwater quality deterioration. 2. Apply hydrogeological and soil science knowledge to assess the impacts of human activities on soil and water resources. 3. Develop skills in mapping and interpreting hydrogeological and soil data to support environmental planning and resource management. 4. Evaluate sustainable practices and policies for preserving natural resources. 5. Demonstrate proficiency in communicating geological concepts and solutions to diverse stakeholders, promoting informed decision-making in environmental management.			
Course Title	ESMJ84: Remote Sensing and GIS			
Category of Course	Major			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	02	---	02
	Hour of Teaching (Total)	30	---	30
Course Objectives (not more than 100 words)	To understand the principles, applications, trends, and pertinent issues of geographical information systems and sciences, including remote sensing (RS), cartography, and Global positioning systems (GPS).			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Electromagnetic Radiation – characteristics, remote sensing regions, and bands; general characteristics of remote sensing satellites, its orbital parameters, and sensors; natural objects spectra – soil, rock, water, and vegetation.			06
II	Digital Image Processing – characteristics of remote sensing data, georeferencing, preprocessing/correction, enhancements, and classification; elements of photo and imagery pattern and interpretation; application in geology.			09
III	Hyperspectral, Thermal, and Microwave Remote Sensing; remote sensing applications in interpreting structure and tectonics, lithological mapping, mineral resources, groundwater potentials, and environmental monitoring.			09
IV	Concept of Datum and Types; different map projection systems; principles and components of GIS; remote sensing data integration with GIS; applications of GIS in various geological studies.			06
Texts / References	1. Drury, S.A. (2001): Image Interpretation in Geology, Allen and Unwin. 2. Gupta, R.P. (1991): Remote Sensing Geology, Springer-Verlag. 3. Lillesand, T.M. and Kiefer, R.W. (1987): Remote Sensing and Image Interpretation, John Wiley. 4. B.S. and Gillespie, A.R. (1980): Remote Sensing in Geology, John Wiley. 5. Burrough, P.A. and McDonnell, R.A. (1998): Principles of Geographic Information Systems.			
Learning Outcomes (not more than 100 words)	The course enhances technical skills and competence in data and information acquisition, extraction, management and analysis; spatial and statistical modelling; mapping and visualization. Further, the course helps develop applications of environmental remote sensing and GIS which can directly enhance service delivery on geology, land use management, ground water management/prospects, agriculture, forestry, food and water security, disaster management, etc.			

Course Title		ESMJ85: Basin Analysis				
Category of Course		Major				
Credits& Hour of Teaching			Theory	Practical	Cumulative	
		Credits	02	---	02	
		Hour of Teaching (Total)	30	---	30	
Course Objectives (not more than 100 words)		To develop concept of the basin analysis. Moreover, the students will acquire knowledge about the various tools involved in basin analysis. They will get firsthand knowledge about the allogenic and autogenic controls and evolution of the sedimentary basins. Furthermore, they will study details of the facies analysis, depositional environments and sequence stratigraphy, besides Quaternary Sedimentology.				
Course Content		The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content					Hr. of Teaching
I	Principles of basin analysis; Methods and techniques of basin analysis; Tectonic classification of sedimentary basins, factors and processes controlling sedimentation; Sedimentary basins and their geothermal evolution.					10
II	Facies analysis, sedimentary facies and facies models with Indian examples; Sediment dispersal patterns and paleocurrent analysis. Depositional environments of different sedimentary basins such as fluvial, shallow marine, and deep					10
III	Facies analysis, sedimentary facies and facies models with Indian examples; Sediment dispersal patterns and paleocurrent analysis. Depositional environments of different sedimentary basins such as fluvial, shallow marine, and deep					10
Texts / References		1. Allen, P.A., and Allen, J.R.L. (2005): Basin Analysis: Principles and Application, Blackwell Publishing. 2. Miall, A.D. (2000): Principles of Basin Analysis, Springer-Verlag. 3. Perry, C.T., and Taylor, K.G. (2006): Environmental Sedimentology, Blackwell Publishing, U.K. 4. Reading, H.G. (1996): Sedimentary Environments and Facies, Blackwell Scientific Publishing. 5. Reineck, H.E., and Singh, I.B. (1978): Depositional Sedimentary Environments, Springer-Verlag.				
Learning Outcomes (not more than 100 words)		This is an applied course that is related to hydrocarbon industry. In this course, the students will acquire knowledge about the various tools involved in basin analysis. They will get knowledge about the allogenic and autogenic controls and tectonic evolution of the sedimentary basins, facies analysis, sequence stratigraphy and Quaternary Sedimentology.				

Course Title	ESMJ86: Practical connected with ESMJ81				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	---	01	01	
	Hour of Teaching (Total)	---	30	30	
Course Objectives (not more than 100 words)	This course aims to study the identification of various microfossils based on their morphology. It also includes the study of sea floor sediment and bottom topography, as well as laboratory techniques for sample processing. Additionally, the course covers various computer applications used in this field of study.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Types of microfossils- calcareous, siliceous, phosphatic and organic walled microfossils.				14

II	Study of major sea-floor features.Study of ocean floor sediments/ rocks/ minerals.	6
III	Grain size analysis of ocean floor sediments.Use of Ocean Data view in location plot and data visualization	10
Texts / References	1. Pratul Kumar Saraswati and M.S. Srinivasan (2016): Micropaleontology: Principles and Applications, Springer, Cham, Heidelberg, New York, Dordrecht, London. 2. Kennett, James P. (1982): Marine Geology, Prentice-Hall, Englewood Cliffs, N.J. 3. Burk, C.A. and Drake, C.L. (2013): The Geology of Continental Margins, Springer.	
Learning Outcomes (not more than 100 words)	By the end of this course, students will be able to identify various microfossils and understand various seafloor features. They will be proficient infield techniques for data and sample collections, laboratory techniques for sample processing and skilled in using computer applications relevant to oceanographic studies, enabling them to conduct comprehensive research in marine geology and climatology.	

Course Title	ESMJ87: Practical connected with ESMJ82				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits		01	01	
	Hour of Teaching (Total)		30	30	
Course Objectives (not more than 100 words)	This course is ideal for newcomers to learn about natural resources and their practical assessment. Practical knowledge to strategic analysis for mineral resource classification and their economic evaluations. To develop scientific and technical knowledge of report writing on mineral policy.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Study of United Nations Framework Classification (UNFC) for Georesources as detailed in the theory syllabus.				05
II	Mineral Resource Valuation Techniques: Cur-off Grades, Tonnage Factor, Specifications, Reserve Estimation.				10
III	Analysis and graphical representation of important Georesources. Preparation of maps showing distribution of important Georesources of India and World.				10
IV	Learning and report writing on national and international mineral policy.				05
Texts / References	1. R.K. Sinha and N.L. Sharma (2019): Mineral Economics, 4th Edition, Oxford & IBH Publishing, pp. 410. 2. G.S. Roonwal (2018): Mineral Exploration: Practical Application, Springer Nature Singapore, pp. 1-298. 3. J.H. Reedman (1979): Techniques in Mineral Exploration, Applied Science Publishers Ltd., London, pp. 1-533. 4. M.E. Rossi and C.V. Deutsch (2014): Mineral Resource Estimation, Springer Dordrecht Heidelberg, pp. 1-332. 5. Haddon F. King (1982): A Guide to the Understanding of Ore Reserve Estimation, Australasian Institute of Mining and Metallurgy, Australia, pp. 1-21. 6. Indian Minerals Yearbook, Indian Bureau of Mines, Ministry of Mines, Government of India. 7. National Mineral Policy, Ministry of Mines. https://mines.gov.in 8. United Nations Framework Classification for Fossil Energy and Mineral Reserves and Resources.				
Learning Outcomes (not more than 100 words)	Demonstrate understanding of the position of the minerals industry in the Global economy. Deliberate the essential concepts of mineral reserve estimation using geoscientific methods, including the concepts of ore reserve estimation. Apply the knowledge gained to understanding, reviewing, and performing mineral reserve estimations.				

Course Title	ESMJ88: Practical connected with ESMJ83				
Category of Course	Major				
Credits& Hour of Teaching		Theory	Practical	Cumulative	
	Credits	---	01	01	
	Hour of Teaching (Total)	---	30	30	
Course Objectives (not more than 100 words)	The Environmental Geology and Hydrogeology theory course aims to provide students with a comprehensive understanding of the Earth's geological processes and their impact on water resources. Objectives include developing knowledge of soil and rock formation, groundwater flow dynamics, and the interactions between geological formations and water systems; understanding the principles of aquifer properties and groundwater contamination. Develop skill to assess the environmental impact of geological processes and human activities on water resources; and equipping students with the skills to apply geological and hydrogeological concepts in environmental assessments, resource management, and sustainability practices. This course prepares students for advanced studies and careers in hydrogeology and environmental management.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
I	Water balance computation of a micro watershed: Preparation and interpretation of water level contour maps and depth to water level maps. Delineation of hydrological boundaries on water table contour maps and analysis of hydrographs for differing groundwater conditions. Estimation of aquifer properties such as hydraulic conductivity, storage coefficient, and transmissivity. Analysis of hydrographs for various components				15
II	Physico-chemical analysis of groundwater and graphical presentation of water quality: Determination of groundwater suitability for drinking and irrigation purposes. Simple numerical problems related to: determination of permeability in field and laboratory, groundwater flow, well hydraulics, etc.				15
Texts / References	1. Todd, D. K. (2006). Groundwater Hydrology, 2nd Ed., John Wiley & Sons, N.Y. 2. Davis, S. N., and DeWeist, R. J. M. (1966). Hydrogeology, John Wiley & Sons Inc., N.Y. 3. Karanth, K. R. (1987). Groundwater: Assessment, Development and Management, Tata McGraw-Hill Pub. Co. Ltd.				
Learning Outcomes (not more than 100 words)	Upon completing the Environmental Geology and Hydrogeology Practical course, students will: 1. Ability to interpret hydrogeological data and maps to assess the impact of human activities on water resources. 2. Gain knowledge of groundwater occurrence, distribution, and movement through analyzing the piezometer data. . 3. Develop data analysis, critical thinking, and problem-solving skills, preparing them to tackle environmental and hydrological issues effectively. Essential for environmental consulting, research, and water resource management careers.				

Course Title	ESMJ89: Practical connected with ESMJ84			
Category of Course	Major			
Credits& Hour of Teaching		Theory	Practical	Cumulative
	Credits	---	01	01
	Hour of Teaching (Total)	---	30	30
Course Objectives (not more than 100 words)	It is to focus on hands-on experience and the application of theoretical knowledge of aerial photographs, satellite images and GIS to real-world scenarios. 1. Learn to use aerial photos and satellite images for geological interpretation. 2. Learn to integrate remote sensing data with other GIS data sources. 3. Understand the process of georeferencing and digitizing spatial data.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			

Units	Course Content	Hr. of Teaching
I	Geomorphology through aerial photos and satellite imagery; Terrain aspect mapping; Determination of scale in aerial photos, measurement of heights of objects from aerial photos, study and interpretation of stereo pair aerial photos; Preparation of interpretation keys, thematic mapping from aerial photos and satellite images – structure, lithology, landforms, minerals, soils, groundwater; Application of GIS in geological studies.	30
Texts / References	1. Drury, S. A. (2001). Image Interpretation in Geology, Allen and Unwin. 2. Gupta, R. P. (1991). Remote Sensing Geology, Springer-Verlag. 3. Lillesand, T. M., and Kiefer, R. W. (1987). Remote Sensing and Image Interpretation, John Wiley. 4. B. S. and Gillespie, A. R. (1980). Remote Sensing in Geology, John Wiley. 5. Burrough, P. A., and McDonnell, R. A. (1998). Principles of Geographic Information Systems	
Learning Outcomes (not more than 100 words)	Students will gain practical skills and knowledge in handling aerial photos, satellite images, and GIS, enabling them to effectively use these technologies in various professional applications.	

Course Title	ESMJ810: Practical connected with ESMJ85				
Category of Course	Major				
Credits & Hour of Teaching		Theory	Practical	Cumulative	
	Credits	---	01	01	
	Hour of Teaching (Total)	---	30	30	
Course Objectives (not more than 100 words)	This course is designed to impart the practical knowledge regarding the basin analysis. Here, the students are trained to use various tools for basin analysis. This also involves the students to interpret various seismic profiles in a sequence stratigraphic framework.				
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.				
Units	Course Content				Hr. of Teaching
	• Study of isopach and facies maps. • Study of facies assemblages of different sedimentary environments. • Preparation of lithologs from facies maps. • Correlation of lithological profiles of important Indian sedimentary basins. • Preparation of rose diagrams and statistical analysis of the paleocurrent data. • Detailed study of seismic profiles in a sequence stratigraphic framework.				30
Texts / References	1. Perry, C. T. and Taylor, K. G. (2006): Environmental Sedimentology, Blackwell Publ., U.K. 2. Reading, H. G. (1996): Sedimentary Environments and Facies, Blackwell Scientific Publ. 3. Emery, D. and Myers, K. (1996): Sequence Stratigraphy, Blackwell Science Ltd. 4. Haq, B. U. (1995): Sequence Stratigraphy and Depositional Response to Eustatic, Tectonic, and Climatic Forcing, Springer.				
Learning Outcomes (not more than 100 words)	The students will learn about the various tools involved in the basin analysis. They will learn to interpret the lithologs and seismic profiles that can be used for petroleum exploration.				

Course Title	Same as major ESMJ84 and ESMJ89			
Category of Course	Minor			
Credits& Hour of Teaching		Theory	Practical	Cumulative
	Credits	03	01	04
	Hour of Teaching (Total)	45	30	75

Research Students

Semester: VII -R

Same as VII-H

Semester: VIII –R

<i>Any one major elective courses from ESMJ81 to ESMJ85 along with connected practical (4 credit each)(List Same as VIII-H)</i>
ES-Minor: same as major ESMJ84 and ESMJ89*
ES-DS8R1: Dissertation

Course Title	ES-DS8R1: Dissertation			
Category of Course	Major			
Credits& Hour of Teaching		Theory	Practical	Cumulative
	Credits	---	12	12
	Hour of Teaching (Total)	---	360	360
Course Objectives (not more than 100 words)	Aims to contribute new knowledge or insights to the field of Earth Science. Here are some common objectives for such research: 1. To deepen the understanding of geological processes, structures, or phenomena. 2. To address a particular problem or question in geology 3. To develop or refine methodologies for geological investigation. 4. To apply geological knowledge to practical issues, such as natural resource management, environmental protection, or geohazard mitigation. 5. To test specific hypotheses related to geological theories or models, using empirical data or simulations to validate or refute them. 6. To provide new or updated geological maps that improves the understanding of geological formations and structures in a specific region. 7. To document and analyze the geological history of an area, providing insights into past climates, tectonic events, or biological evolution.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Research work under supervision of a research guide is to be submitted in the form of a dissertation			360
Texts / References				
Learning Outcomes (not more than 100 words)	The learning outcomes of a research dissertation in geology typically reflect both the specific achievements of the project and the broader skills developed during the research process. Advanced understanding of geological concepts, theories, and processes relevant to your research topic. Awareness of current research trends, techniques, and findings in the field of geology.			